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Latency and Underreporting in Cholera Transmission: A Structural Robustness Analysis of the Control Reproduction Number R_c (West Africa)

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Abstract

This thesis is part of the fight against cholera in West Africa, a region marked by recurrent epidemics linked to poor sanitary conditions and environmental factors that favor the persistence of the pathogen *Vibrio cholerae*. In order to improve the understanding and prediction of this epidemic dynamic, a compartmental SEIR-B model (Susceptible–Exposed–Infectious–Recovered with bacterial reservoir) was developed. This model integrates not only human and environmental transmission pathways but also clinical and surveillance heterogeneities, distinguishing between asymptomatic, symptomatic, reported, and unreported cases.

The central focus of this work is the estimation and robustness analysis of the control reproduction number (R_c), a key epidemiological indicator for determining intervention thresholds. Using the Next Generation Matrix (NGM) method, R_c was decomposed into specific contributions from each transmission pathway, thus identifying the most effective control levers (vaccination, improved hygiene, enhanced surveillance).

Two major sources of uncertainty were taken into account:

1. Structural uncertainty related to the exposed compartment (E), whose relevance was evaluated through analytical comparison of models with and without latency.
2. Parametric uncertainty related to chronic underreporting of cases, which biases transmissibility estimates.

A local and global sensitivity analysis (PRCC, Latin Hypercube Sampling) was conducted to identify the parameters that most strongly influence R_c and to assess the robustness of the results in the face of these uncertainties. The results show that the inclusion of the latent compartment significantly modifies the structure of R_c , and that low case detection scenarios substantially increase uncertainty in epidemic projections.

Before these analyses, the mathematical properties of the model (existence, positivity, boundedness, and stability of solutions) were rigorously demonstrated, ensuring the validity of the conclusions.

Résumé

Ce mémoire s'inscrit dans le contexte de la lutte contre le choléra en Afrique de l'Ouest, une région marquée par la récurrence d'épidémies liées à des conditions sanitaires précaires et à des facteurs environnementaux favorisant la persistance du pathogène *Vibrio cholerae*. Afin

d'améliorer la compréhension et la prévision de cette dynamique épidémique, un modèle compartimental de type SEIR-B (Susceptibles–Exposés–Infectieux–Rétablis avec réservoir bactérien) a été développé. Ce modèle intègre non seulement les voies de transmission humaines et environnementales, mais aussi les hétérogénéités cliniques et de surveillance, en distinguant les cas asymptomatiques, symptomatiques, déclarés et non déclarés.

L'axe central de ce travail est l'estimation et l'analyse de robustesse du nombre de reproduction de contrôle (R_c), indicateur épidémiologique clé pour déterminer les seuils d'intervention. En utilisant la méthode de la matrice de nouvelle génération (NGM), le R_c a été décomposé en contributions spécifiques de chaque voie de transmission, permettant ainsi d'identifier les leviers les plus efficaces de contrôle (vaccination, amélioration de l'hygiène, surveillance accrue).

Deux sources majeures d'incertitude ont été prises en compte :

1. L'incertitude structurelle liée au compartiment des exposés (E), dont la pertinence a été évaluée par comparaison analytique des modèles avec et sans latence.
2. L'incertitude paramétrique liée à la sous-déclaration chronique des cas, qui biaise les estimations de transmissibilité.

Une analyse de sensibilité locale et globale (PRCC, Latin Hypercube Sampling) a permis d'identifier les paramètres influençant le plus fortement R_c et d'évaluer la robustesse des résultats face à ces incertitudes. Les résultats montrent que l'inclusion du compartiment latent modifie de manière significative la structure de R_c et que les scénarios de faible détection des cas accroissent substantiellement l'incertitude sur les projections épidémiologiques.

Mais bien avant, les propriétés mathématiques du modèle (existence, positivité, bornitude et stabilité des solutions) ont été rigoureusement démontrées, assurant la validité des conclusions.

Acronyms

Table 1: List of acronyms

Acronym	Full form	Short description / notes
ODE	Ordinary Differential Equation	Mathematical equations used to describe model dynamics.
NGM	Next-Generation Matrix	Method for computing reproduction numbers and stability analysis.
DFE	Disease-Free Equilibrium	Equilibrium state with no infection present.
CFEP	Cholera-Free Equilibrium Point	Equilibrium state specific to cholera-free conditions.
R_c	Control reproduction number	Reproduction number under control/intervention measures.
R_0	Basic reproduction number	Basic (uncontrolled) reproduction number.
GSA	Global Sensitivity Analysis	Method to assess global parameter sensitivity.
PRCC	Partial Rank Correlation Coefficient	Sensitivity index used in PRCC analysis.
LHS	Latin Hypercube Sampling	Sampling technique for uncertainty/sensitivity analysis.
WASH	Water, Sanitation and Hygiene	Public health intervention category (water, sanitation, hygiene).
SIQRB	Susceptible–Infected–Quarantined–Recovered–B	Compartmental model cited in references (includes bacterial reservoir).

Contents

viii. Abstract	i
vii. Résumé	i
iii. Acronyms	iii
iv. Table of contents	iv
v. List of tables	v
vi. List of figures	vi
1 Introduction	1
1.1 Literature Review: Methodological Justification	2
1.2 Objectives	4
2 Methodology	5
2.1 Model formulation	5
2.2 Bifurcation	14
2.3 Comparison strategy	15
2.4 Sensitivity analysis: Detection scenarios	16
3 Results	18
3.1 Model analysis	18
3.2 Bifurcation Analysis	35
3.3 Numerical Bifurcation Analysis	44
3.4 Comparison of model	49
3.5 Sensitivity Analysis and Structural Robustness Results	53
4 Discussion	56
5 Conclusion	60

List of Tables

1	List of acronyms	iii
2.1	Model Variables	5
2.2	Model Parameters and Units	8
2.3	Baseline parameter values and ranges used in the numerical simulations. All rates are expressed in day^{-1}	13
3.1	Coefficient $a(\varphi)$ for different durations of natural immunity.	45
3.2	Decomposition of $R_c = R_s + R_a + R_r + R_u + R_e$ into five routes.	49

List of Figures

2.1	Flow diagram of the compartmental model of cholera	6
2.2	Exposed (E) compartmentless model flow diagram.	10
3.1	Main bifurcation diagram: I^* (total infectious individuals at equilibrium) as a function of R_c . The endemic branch (solid line) emerges only from $R_c = 1$ onward, with no non-trivial unstable branch below the threshold — the signature of a forward transcritical bifurcation.	46
3.2	Real part of the dominant Jacobian eigenvalue, $\text{Re}(\lambda_{\max})$, as a function of R_c (symlog scale), for the disease-free equilibrium and the endemic equilibrium. The exact crossing at $R_c = 1$ numerically confirms the stability change predicted by center manifold theory.	47
3.3	All equilibrium components (susceptible, vaccinated, exposed, infectious by class, recovered, bacterial reservoir) as a function of R_c . Left: majority compartments S^*, V^* . Right: infectious, recovered, and bacterial reservoir compartments (symlog scale).	47
3.4	Time trajectories $I(t)$ over 30 years (log scale), for $R_c = 0.7$ (below threshold), $R_c = 1.0$ (at threshold), and $R_c = 1.5$ (above threshold), from the same initial perturbation. The $R_c = 1.5$ regime shows damped oscillations before converging to the endemic equilibrium.	48
3.5	Phase portrait ($S/N, I_{\text{total}}$) for $R_c = 0.7$ (left) and $R_c = 1.5$ (right), from several initial conditions. The grey dot marks the starting point, the coral star the equilibrium reached. For $R_c = 1.5$, convergence occurs via a damped spiral, consistent with a pair of complex eigenvalues with negative real part near the endemic equilibrium.	48
3.6	Uncertainty distributions of $\Delta = R_c - R_c^e$ (left) and $\Psi = R_c/R_c^e$ (right), LHS $n = 3000, \pm 20\%$ around the calibrated values.	52
3.7	Local elasticities of R_c (tornado analysis, $\pm 20\%$) for four values of the reporting proportion η	53
3.8	Partial Rank Correlation Coefficients (PRCC) of R_c with respect to the model parameters (LHS, $n = 3000, \pm 20\%$).	54

1. Introduction

Cholera, an acute diarrheal disease caused by *Vibrio cholerae*, remains a global public health concern, but its burden is particularly severe and endemic in West Africa. The region faces a recurring cycle of outbreaks, often seasonal and linked to periods of heavy rainfall [2], and further exacerbated by inadequate sanitation infrastructure and increasing urban population density. Environmental, water-related, and socioeconomic factors together create fertile ground for the disease to persist endemically and spread explosively when conditions allow.

The complexity of cholera transmission in this setting is compounded by its **dual transmission pathway**, combining direct human-to-human contact with indirect, waterborne exposure. As highlighted in foundational research [29], the aquatic environment acts as a critical intermediate reservoir for the pathogen, so that a faithful model must capture not only human epidemiological kinetics but also the ecology of *V. cholerae* in the environment. Understanding and anticipating these regional epidemics accordingly requires mechanistic models sophisticated enough to support effective planning and intervention across West Africa.

In response to this recurring threat, mathematical modeling based on systems of ordinary differential equations (ODEs) has become an essential analytical tool. The SEIR-B model developed in this thesis provides a formal structure for studying the transmission dynamics characteristic of the region.

The central analytical objective is the estimation of the control reproduction number, R_c , which determines the intervention threshold required to halt disease spread. An accurate, structurally decomposed estimate of R_c is essential, as it enables the identification of the most effective levers among vaccination, sanitation improvement, and surveillance strengthening. The Next-Generation Matrix (NGM) method [1] is used here to derive R_c analytically, allowing the specific contribution of each infectious class (asymptomatic, symptomatic, environmental) to be isolated, so that limited public health resources can be directed toward the transmission routes that matter most in this sub-region.

However, the application of these models in West Africa faces major challenges related to operational uncertainty. We identify two critical issues:

1. **Structural uncertainty (latency):** the incubation period, difficult to observe directly in the field, constitutes a structural modeling assumption that, if poorly specified, can bias the estimation of epidemic risk [16].
2. **Parametric uncertainty (differential reporting):** the principal challenge in West

Africa is the limited completeness of surveillance systems. Because of persistent disparities in access to care, a substantial proportion of mild or asymptomatic cases are never reported. Explicitly modeling differential case detection is therefore imperative, in order to assess how uncertainty in the reporting rate η affects the robustness of the R_c estimate and, by extension, the validity of the regional health policies derived from it.

This thesis proposes a structural robustness and sensitivity analysis intended to provide a solid scientific basis, whose conclusions remain valid under structural variation and under the high parametric uncertainty inherent to epidemiological data collected in West Africa.

1.1 Literature Review: Methodological Justification

1.1.1 Modeling Transmission and the Evolution of the Compartmental Structure

Modeling cholera hinges on capturing its **dual transmission dynamics**: an anthropogenic cycle (person-to-person) and an environmental cycle (water-to-person). Compartmental SEIR (Susceptible–Exposed–Infectious–Recovered) models have accordingly been extended to include a water reservoir compartment, a necessity first formalized in the pioneering work of **Codeço (2001) [2]**, which demonstrated that the endemic persistence of cholera cannot be explained without an explicit bacterial-load compartment B – giving rise to the SEIR-B family of models.

Applying these models to the epidemic reality of **West Africa**, however, requires further **disaggregation** of the infectious classes, in view of the region’s clinical and surveillance heterogeneity. Recent literature justifies distinguishing between **asymptomatic** (I_a) and **symptomatic** (I_s) individuals [17], since the two groups exhibit markedly different bacterial shedding rates and transmissibility. Surveillance heterogeneity in underfunded health systems further requires distinguishing between **reported** (I_r) and **unreported** (I_u) cases. The present work builds on this composite architecture – integrating the latent state (E), the environmental dynamics (B), and the disaggregation of infected individuals – on the premise that the model’s **structural fidelity** is a prerequisite for the robustness of its conclusions.

1.1.2 Control Reproduction Number (R_c): Analytical Tool and Structural Decomposition

The **control reproduction number** (R_c) is the cardinal metric of mathematical epidemiology, serving as the disease's **invasion threshold**. For dynamical systems involving several distinct infectious classes – human and environmental alike – the computation of R_c must proceed via the **Next-Generation Matrix (NGM)** method [23]. This technique is essential because it guarantees a rigorous algebraic derivation at the disease-free equilibrium (DFE), explicitly identifying the new infections generated by each class.

The principal methodological advantage of the NGM lies in its capacity to enable a **structural decomposition** of R_c into a sum of route-specific contributions. This decomposition is a **powerful intervention-planning tool** [39], allowing policymakers to precisely target the transmission component whose reduction would most efficiently bring R_c below unity.

1.1.3 Operational Uncertainty in Regional Epidemiology: Underreporting and Latency

Despite the analytical sophistication of the NGM, its practical application is constrained by the **high parametric uncertainty** inherent to field data collection – a limitation that is particularly acute in the West African context.

The mathematical epidemiology literature widely recognizes that models suffer from critical uncertainty stemming from the practical impossibility of precisely measuring every biological parameter [31]. For cholera specifically, this uncertainty is compounded by weaknesses in surveillance systems, most notably in the **case reporting rate**. Dedicated studies on waterborne pathogens, such as [31], have shown that **underreporting** directly affects the estimation of transmissibility (R_c) and may obscure the true epidemic burden. Assessing the sensitivity of R_c to variations in the reporting proportion η is therefore a matter of operational, not merely theoretical, validity.

Moreover, the **latent state** (E) constitutes a source of **structural uncertainty** in its own right. Omitting this compartment, or mis-specifying its residence time, alters the model's internal timing and can significantly bias the resulting calculation of R_c [29] – underscoring the need for an analysis that explicitly evaluates the impact of this particular modeling choice.

1.1.4 Methodological Justification: From Local Analysis to Structural Robustness (GSA)

To move beyond simple point calibration of parameters, this work employs **Global Sensitivity Analysis (GSA)**. Unlike local sensitivity analyses – such as elasticity analysis, which evaluates parameter impact only in the neighborhood of a single point in parameter space – GSA explores the **entire space** of parametric uncertainty to determine which parameters contribute most to the variance of the output (R_c).

The **Partial Rank Correlation Coefficient (PRCC)** [33] is a well-established GSA method for quantifying the monotonic influence of individual parameters. The most significant methodological contribution of this thesis, however, lies in extending GSA to evaluate the model's **structural robustness**: this is achieved by examining how the hierarchy and relative influence of the model's parameters are **reordered** when the model structure itself is modified (model with E versus model without E), or when key operational parameters – such as the reporting proportion η – are varied across plausible scenarios. This level of analysis is essential to ensure that public health policy recommendations for West Africa rest not on a modeling artifact, but on a sufficiently robust mathematical foundation.

1.2 Objectives

Our main objective is to contribute to improving the reliability of epidemiological projections for cholera in the West African context, by conducting a structural robustness analysis of the control reproduction number (R_c).

Our specific objectives are:

1. **Development of the SEIR-B model:** to formalize a system of compartmental ordinary differential equations (ODEs) for cholera, incorporating the environmental reservoir, the latent state, and a disaggregation of infectious classes (I_a, I_s, I_r, I_u).
2. **Analysis of the structural impact of the E compartment:** to derive and compare the analytical expressions of R_c for the model with the latent state (E) and for a simplified model without E , using the NGM, in order to quantify the structural influence of latency on epidemic risk assessment.
3. **Quantification of sensitivity and robustness:** to conduct a Global Sensitivity Analysis (GSA), using PRCC together with a local elasticity analysis, to quantify the sensitivity of R_c to scenarios of differential case detection.

2. Methodology

2.1 Model formulation

We formulate a deterministic human–environment compartmental model for cholera transmission. The human population is partitioned into susceptible S , vaccinated V , exposed E , asymptomatic infectives I_a , symptomatic infectives I_s who are then classified as *reported* I_r or *unreported* I_u , and recovered/immune R . The environmental reservoir B represents the concentration of *V. cholerae* in contaminated water. All compartments are time-dependent, $\{S(t), V(t), E(t), I_a(t), I_s(t), I_r(t), I_u(t), R(t)\}$ for humans and $B(t)$ for the environment [3, 4].

The table below lists the parameters used in the cholera transmission model, along with their notation, baseline values, units, and sources as reported by [3] :

Table 2.1: Model Variables	
Variable	Description
Human compartments	
S	Susceptible Human
V	Vaccinated Human
E	Exposed Human
I_a	Infected asymptomatic Human
I_s	Infected symptomatic Human
I_r	Infected symptomatic reported Human
I_u	Infected symptomatic undetected Human
R	Recovery Human
Environmental compartment	
B	<i>Vibrio cholerae</i>

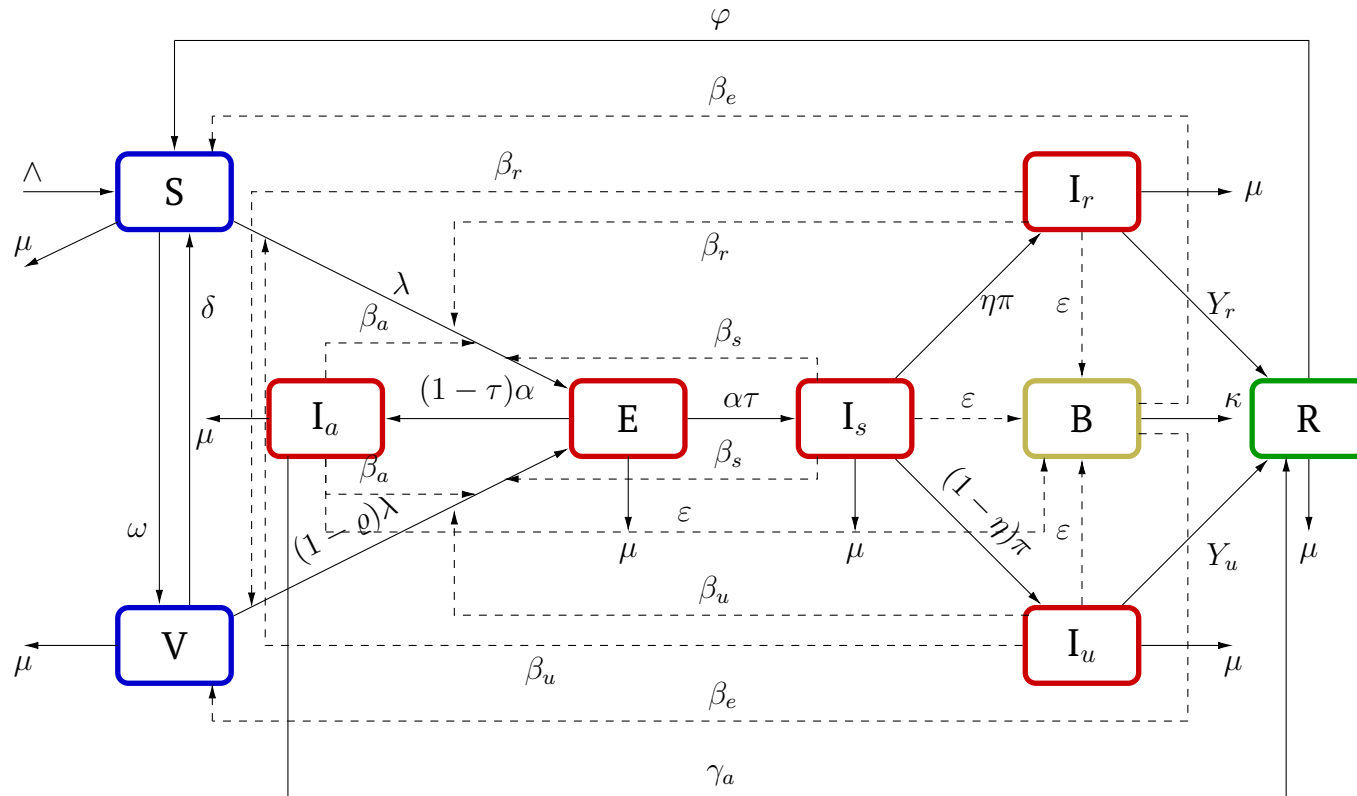


Figure 2.1: Flow diagram of the compartmental model of cholera transmission with bacterial reservoir and control measures.

Individuals are recruited into the susceptible class S at rate Λ . All human compartments are subject to natural mortality at rate μ . Susceptibles may be vaccinated at rate ω into V ; vaccine protection is leaky, so vaccinated individuals become infected at rate $(1 - \varrho)\lambda(t)$. Immunity wanes at rate δ from $V \rightarrow S$ (vaccine-derived) and at rate φ from $R \rightarrow S$ (infection-derived).

The *force of infection* combines person-to-person and water-mediated transmission,

$$\lambda(t) = \frac{(\beta_a I_a(t) + \beta_s I_s(t) + \beta_r I_r(t) + \beta_u I_u(t))}{N} + \frac{\beta_e B(t)}{K + B(t)}, \quad (2.1.1)$$

where $\beta_\bullet$ are transmission coefficients and K is a half-saturation constant. Newly infected individuals enter E ; after an average incubation $1/\alpha$, a fraction τ become symptomatic I_s and the remainder $1 - \tau$ asymptomatic I_a . Symptomatic cases are classified at rate π into reported I_r (fraction η) and unreported I_u (fraction $1 - \eta$). In the analytical analysis we consider $\beta_\bullet/N$ as $\beta_\bullet$ to simplify the different expression. Infectious classes recover to R at rates $\gamma_a, \gamma_r, \gamma_u$. All infectious classes shed bacteria into the environment at a per-capita rate ε ; the environmental concentration B decays at rate κ .

The mathematical expression of the cholera dynamics compartmental model (Figure 2.1) is given by the system of equations below.

$$\left\{ \begin{array}{l} \frac{dS}{dt} = \Lambda - (\lambda + \omega + \mu)S + \delta V + \varphi R, \\ \frac{dV}{dt} = \omega S - [(1 - \varrho)\lambda + \delta + \mu]V, \\ \frac{dE}{dt} = \lambda S + (1 - \varrho)\lambda V - (\alpha + \mu)E, \\ \frac{dI_a}{dt} = (1 - \tau)\alpha E - (\gamma_a + \mu)I_a, \\ \frac{dI_s}{dt} = \tau\alpha E - (\pi + \mu)I_s, \\ \frac{dI_r}{dt} = \eta\pi I_s - (\gamma_r + \mu)I_r, \\ \frac{dI_u}{dt} = (1 - \eta)\pi I_s - (\gamma_u + \mu)I_u, \\ \frac{dR}{dt} = \gamma_a I_a + \gamma_r I_r + \gamma_u I_u - (\mu + \varphi)R, \\ \frac{dB}{dt} = \varepsilon(I_a + I_s + I_r + I_u) - \kappa B, \end{array} \right. \quad (2.1.2)$$

with $\lambda(t)$ given by (2.1.1).

Initial conditions and feasible region. Assume nonnegative initial data

$$S(0) = S_0, V(0) = V_0, E(0) = E_0, I_a(0) = I_{a0},$$

$$I_s(0) = I_{s0}, I_r(0) = I_{r0}, I_u(0) = I_{u0}, R(0) = R_0, B(0) = B_0 \geq 0.$$

Let us pose, $\mathcal{C}(t) = (S, V, E, I_a, I_s, I_r, I_u, R, B)$ and $N = S + V + E + I_a + I_s + I_r + I_u + R$.

Finally, during the study period, the population is considered closed, that is, there is no immigration or emigration. This assumption is consistent with short-term modeling and appropriate for a disease with a low mortality rate, such as cholera[17].

Table 2.2: Model Parameters and Units

Parameter	Description	Unit
Demographic parameters		
Λ	Recruitment rate of susceptible population	individuals/day
μ	Natural death rate of human	day ⁻¹
ω	Vaccination rate	day ⁻¹
δ	Rate of loss of immunity after vaccination	day ⁻¹
Disease progression		
α	Incubation progression rate	day ⁻¹
τ	Probability of exposed to be infected symptomatic	–
$1 - \tau$	Probability of exposed to be infected asymptomatic	–
π	Detection progression rate	day ⁻¹
η	Rate of infected symptomatic to be detected	–
$1 - \eta$	Rate of infected symptomatic to be undetected	–
λ	Force of infection	day ⁻¹
Recovery and immunity		
γ_a	Infected asymptomatic spontaneous recovery rate	day ⁻¹
γ_r	Infected reported spontaneous recovery rate	day ⁻¹
γ_u	Infected unreported spontaneous recovery rate	day ⁻¹
ϱ	Vaccine efficacy	–
Transmission parameters		

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Table 2.2 continued from previous page

Parameter	Description	Unit
β_a	Transmission rate of S and V individuals by the asymptomatic	$\text{ind}^{-1}\text{day}^{-1}$
β_s	Transmission rate of S and V individuals by the symptomatic	$\text{ind}^{-1}\text{day}^{-1}$
β_r	Transmission rate of S and V individuals by the infected reported	$\text{ind}^{-1}\text{day}^{-1}$
β_u	Transmission rate of S and V individuals by the infected undetected	$\text{ind}^{-1}\text{day}^{-1}$
Environmental parameters		
ε	Contribution of infected individuals to the population of <i>Vibrio cholerae</i>	–
κ	Net death rate of <i>Vibrio cholerae</i>	day^{-1}
β_e	Rates of ingesting vibrios from the contaminated environment	day^{-1}

2.1.1 Assessing the relevance of the exposed compartment via reproduction-number comparison

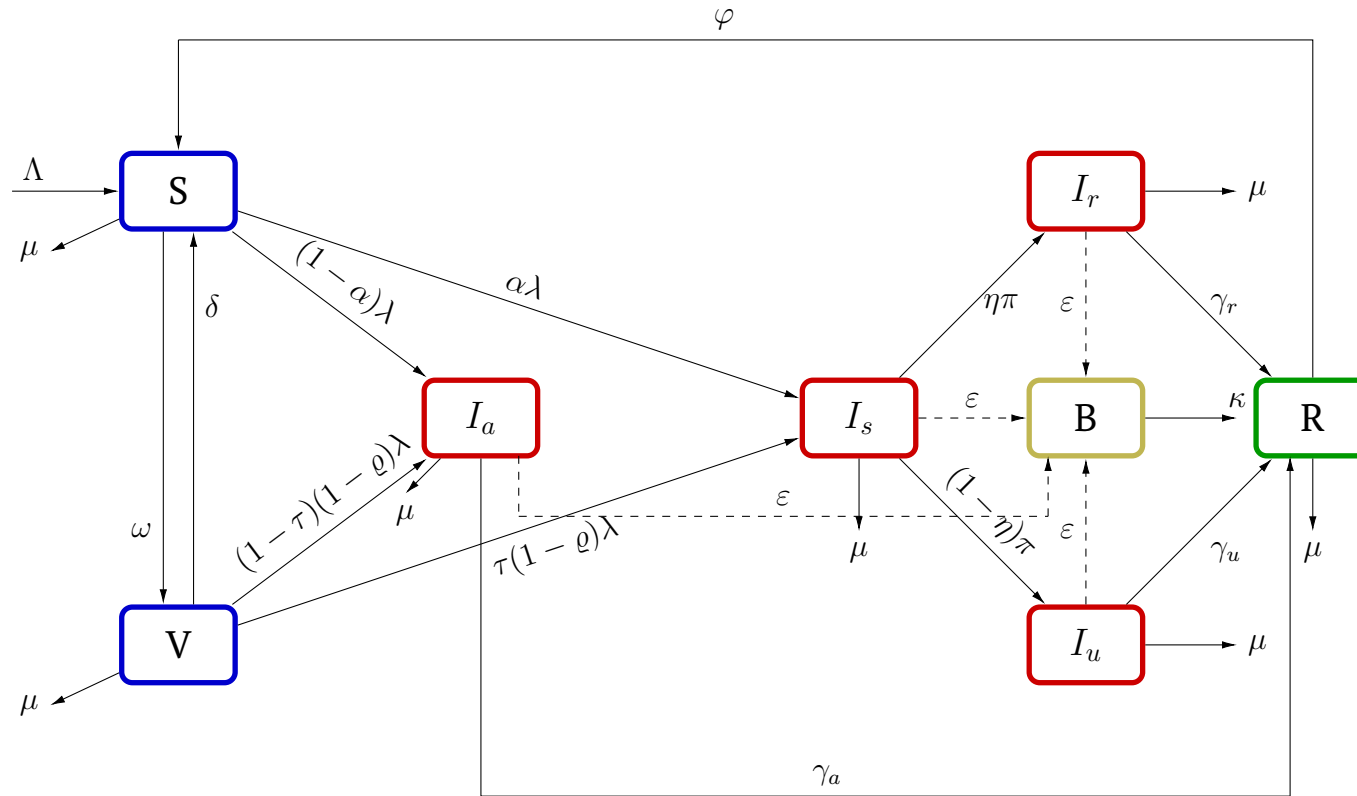


Figure 2.2: Exposed (E) compartmentless model flow diagram.

We assess whether an explicitly exposed compartment E is warranted by comparing the control reproduction number R_c of two model variants. In both cases, R_c is computed at the disease-free equilibrium (DFE) using the next-generation matrix (NGM) approach [1].

We adopt the standard incidence combining human infectious classes and an environmental reservoir B :

The dynamics read

$$\left\{ \begin{array}{l} \frac{dS}{dt} = \Lambda + \varphi R + \delta V - \lambda S - \delta S - \mu S, \\ \frac{dV}{dt} = \omega S - \delta V - (1 - \varrho)\lambda V - \mu V, \\ \frac{dI_a}{dt} = (1 - \alpha)\lambda S + (1 - \tau)(1 - \varrho)\lambda V - (\mu + \gamma_a)I_a, \\ \frac{dI_s}{dt} = \alpha\lambda S + \tau(1 - \varrho)\lambda V - (\pi + \mu)I_s, \\ \frac{dI_r}{dt} = \pi\eta I_s - (\gamma_r + \mu)I_r, \\ \frac{dI_u}{dt} = \pi(1 - \eta)I_s - (\gamma_u + \mu)I_u, \\ \frac{dR}{dt} = \gamma_a I_a + \gamma_r I_r + \gamma_u I_u - (\mu + \varphi)R, \\ \frac{dB}{dt} = \varepsilon(I_a + I_s + I_r + I_u) - \kappa B. \end{array} \right. \quad (2.1.3)$$

Setting

$$N_e = S + V + I_a + I_s + I_r + I_u + R, \quad I_a = I_s = I_r = I_u = R = B = 0 \text{ hence } \lambda = 0 \text{ as DEF}$$

The control reproduction number is the spectral radius denoted as R_c^e .

In the canonical case where E is not infectious, the latent stage often multiplies the effective transmission term by $\nu/(\nu + \mu_E) \leq 1$, reducing R_c due to losses during incubation [8]. The environmental pathway B is modeled as in waterborne diseases (e.g., cholera) [2].

2.1.2 Data

WHO surveillance aggregates (cholera_adm0_public_2024.csv) for four West African countries (Nigeria, Ghana, Niger, Togo), totaling 30,807 cases over a 51-week epidemiological window (January 1 – December 23, 2024).

UN World Population Prospects indicators (total deaths, crude death rate) for the 16 ECOWAS + Mauritania countries, used to estimate μ (regional natural mortality) and a reference demographic population.

The available files provide only aggregated annual totals, not a weekly incidence series. A synthetic weekly curve (rapid log-normal rise + endemic tail) was constructed, constrained so that its cumulative sum exactly matches the real observed total (30,807 cases) over the real observed duration (51 weeks). This curve serves as the calibration target, explicitly documented as a constructed proxy rather than raw data.

2.1.3 Identifiability Problem and Effective Population

A preliminary diagnostic showed that with N = total regional population ($\sim 4.6 \times 10^8$), the model systematically produces millions of cases as soon as $R_c > 1$, regardless of the β value used (even $\sim 10^{-4}$), whereas the observed outbreak counted only 30,807 cases (attack rate $\sim 7 \times 10^{-5}$). This is a well-documented **structural identifiability** problem in cholera modeling [2]: a real outbreak does not mix an entire national population, but a localized “effective” population (urban foci, flood-affected areas).

An N_{eff} parameter is therefore introduced and *calibrated*, distinct from the demographic population N (used only to fix μ). N_{eff} replaces N as the denominator of λ and the size of the initial susceptible pool, with its own local demographic renewal ($\Lambda_{\text{eff}} = \mu \cdot N_{\text{eff}}$), bounded to $[5 \times 10^4, 5 \times 10^7]$ individuals.

2.1.4 Constrained Least-Squares Calibration

The five transmission parameters and N_{eff} are estimated using `scipy.optimize.least_squares` (Trust Region Reflective algorithm) on the target weekly series, with:

Hierarchical constraints are imposed *by construction*, not merely checked post hoc, via a free vector $x \in \mathbb{R}^6$:

$$\beta_a = e^{x_0}, \quad \beta_s = \beta_a + e^{x_1} \quad (\Rightarrow \beta_s > \beta_a), \quad (2.1.4)$$

$$\beta_u = e^{x_2}, \quad \beta_r = \beta_u \cdot \sigma(x_3) \quad (\Rightarrow 0 < \beta_r < \beta_u), \quad (2.1.5)$$

$$\beta_e = e^{x_4}, \quad N_{\text{eff}} = e^{x_5} \text{ (bounded)}, \quad (2.1.6)$$

where σ is the logistic function. These constraints reflect: $\beta_s > \beta_a$ (symptomatic individuals shed more than asymptomatic ones) and $\beta_r < \beta_u$ (reported cases, isolated/treated, transmit less than unreported cases circulating freely).

Anscombe (variance-stabilizing) transform, $\sqrt{y + 3/8}$, appropriate for Poisson-like count

data.

A single aggregated incidence curve cannot, on its own, identify five separate transmission routes, a classical structural identifiability problem in ODE calibration. Without additional constraints, the optimizer pushes human β 's toward zero and assigns the entire fit to β_e . A light ridge regularization term (weight $\lambda_{\text{reg}} = 0.15 \ll$ the data-fit weight) is added, centered on literature-informed *relative ratios* (not absolute values), a maximum-a-posteriori (MAP) estimator with a weakly informative prior.

Six independent optimizations from perturbed initial points, to guard against local minima.

Table 2.3: Baseline parameter values and ranges used in the numerical simulations. All rates are expressed in day^{-1} .

Parameter	Baseline	Range	Description	Source
Demographic parameters				
N	4.562×10^8	–	Total regional population (West Africa, UN WPP subregion, 2024)	[37]
N_{eff}	3.644×10^5	$[5 \times 10^4, 5 \times 10^7]$	Effective outbreak-mixing population (localized transmission pool)	This study (calibrated)
Λ	9.71	–	Recruitment rate ($\Lambda = \mu N_{\text{eff}}$)	This study (derived)
μ	2.664×10^{-5}	–	Natural mortality rate	[37]
ω	2.74×10^{-4}	$(0-5 \times 10^{-3})$	Vaccination rate	Assumed
δ	9.13×10^{-4}	$(5 \times 10^{-4}, 2 \times 10^{-3})$	Vaccine waning rate	[10]
Disease progression				
α	0.7143	–	Incubation rate	[13]
τ	0.25	$(0.20, 0.30)$	Fraction developing symptoms	[9]
π	0.20	$(0.10, 0.30)$	Progression rate from symptomatic infection	Assumed
η	0.30	$(0.05, 0.50)$	Reporting proportion	Assumed
Recovery and immunity				
γ_a	0.15	–	Recovery rate of asymptomatic individuals	[14]
γ_r	0.25	–	Recovery rate of reported cases	[14]

Continued on next page

Table 2.3 continued from previous page

Parameter	Baseline	Range	Description	Source
γ_u	0.20	–	Recovery rate of unreported cases	[11]
φ	9.13×10^{-4}	$(0, 5 \times 10^{-2})$	Natural immunity waning rate	[39]
ϱ	0.86	$(0.70, 0.95)$	Vaccine efficacy	[14]
Transmission parameters (calibrated)				
β_a	4.546×10^{-4}	–	Transmission coefficient (asymptomatic)	(calibrated)
β_s	4.543×10^{-3}	–	Transmission coefficient (symptomatic)	(calibrated)
β_r	3.171×10^{-4}	–	Transmission coefficient (re-reported)	(calibrated)
β_u	1.449×10^{-3}	–	Transmission coefficient (unreported)	(calibrated)
β_e	6.075×10^{-2}	–	Environmental transmission coefficient	(calibrated)
Environmental parameters				
ε	13	–	Bacterial shedding rate	[40]
κ	0.3333	–	Bacterial decay rate	[40]
K	10^6	–	Half-saturation constant	[2]

2.2 Bifurcation

The controlled reproduction number R_c determines, to first order, the local stability of the disease-free equilibrium (DFE): it is locally asymptotically stable for $R_c < 1$ and unstable for $R_c > 1$. This linear result, however, says nothing about the possible existence of an endemic equilibrium for $R_c < 1$, nor about the nature of the bifurcation at $R_c = 1$. To settle this question, we apply the center manifold theorem of Castillo-Chavez and Song [?], which allows the direction of the transcritical bifurcation at the critical point to be determined analytically.

The method consists in rewriting the system near the critical point $\beta_s = \beta_s^*$ (defined by $R_c(\beta_s^*) = 1$), computing the right and left eigenvectors associated with the zero eigenvalue

of the Jacobian at this point, and then evaluating the two coefficients of the normal form on the center manifold:

$$a = \sum_{k,i,j=1}^n v_k w_i w_j \frac{\partial^2 f_k}{\partial x_i \partial x_j}(0,0), \quad b = \sum_{k,i=1}^n v_k w_i \frac{\partial^2 f_k}{\partial x_i \partial \beta_s}(0,0), \quad (2.2.1)$$

where w and v denote, respectively, the normalized right and left eigenvectors ($v \cdot w = 1$), evaluated at the disease-free equilibrium and at the critical point β_s^* . The sign of a then determines the direction of the bifurcation:

- $a < 0$: *forward* bifurcation (classical transcritical case) — the endemic equilibrium exists and is stable only for $R_c > 1$, near the threshold;
- $a > 0$ (with $b > 0$): *backward* bifurcation — a stable endemic equilibrium coexists with the stable DFE over an interval $R_c < 1$, meaning that bringing R_c below 1 is no longer sufficient to guarantee elimination.

In our model, the natural bifurcation parameter is the immunity-waning rate φ (the inverse of the mean duration of post-infection protection), which enters linearly into the center manifold coefficients. We can therefore express a as a function of φ and determine, if it exists, a critical threshold φ_c separating a forward regime from a backward regime.

2.3 Comparison strategy

Comparison strategy and decision rule. We compute R_c^e and R_c under identical transmission, recovery, vaccination, and environmental parameters, and report

$$\Delta = R_c - R_c^e, \quad \Psi = \frac{R_c}{R_c^e}. \quad (8)$$

We define the absolute effect of including E on the reproduction number as $\Delta = R_c - R_c^e$. Its sign and magnitude summarize whether E decreases ($\Delta < 0$) or increases ($\Delta > 0$) net transmissibility. Near the invasion threshold, Δ is decision-relevant: if $R_c^e > 1$ but $R_c < 1$ (or vice versa), the exposed stage is pivotal and should be retained. We quantify robustness by constructing an uncertainty interval for Δ (e.g., via Latin Hypercube Sampling): a 95% CI not covering 0 indicates a structurally relevant effect of E ; otherwise, parsimony favors the simpler model.

As a scale-free complement, we also report the ratio $\Psi = R_c/R_c^e$. Interpretation: $\Psi < 1$ indicates that E decreases net transmissibility (non-infectious latency, additional losses); $\Psi > 1$ suggests structural relevance of E (e.g., re-routing toward more transmissive classes

or interaction with the environmental route). To assess robustness, we perform uncertainty analysis via Latin Hypercube Sampling (LHS) and summarize medians/95% CIs of R_c^e , R_c , and Ψ , as well as $\Pr(R_c > R_c^e)$ [18, 19]. Finally, we decompose the R_c and R_c^e to quantify the contributions of the I_a , I_s , I_r , I_u and B . A substantial and robust difference (95% CI of Ψ excluding 1) and/or a marked reallocation of pathway contributions supports inclusion of E will be interpreted.

2.4 Sensitivity analysis: Detection scenarios

Local (elasticity) sensitivity and tornado diagrams. For each parameter $p \in \Theta$, the *normalised (local) elasticity* of R_c is defined as

$$S_p^{R_c} = \frac{\partial R_c}{\partial p} \frac{p}{R_c} = p \frac{\partial}{\partial p} (\ln R_c),$$

which we compute by *direct symbolic differentiation* of the explicit formula of R_c [18, 19]. To study how detection (reporting) modifies the influence of the other parameters, elasticities are evaluated on four fixed detection scenarios $\eta \in \{0.05, 0.20, 0.35, 0.50\}$ while keeping all other parameters at their references. For each η , we rank the magnitudes $|S_p^{R_c}|$ and present the results as *tornado charts* (Fig. 3.7; panels a–d), with green (resp. red) bars for positive (resp. negative) contributions. No numerical values are reported in the text, as the figures provide the full quantitative read–out used in the discussion.

Global (PRCC) sensitivity. Global influence is assessed with *partial rank correlation coefficients* (PRCC). We generate N parameter sets by *Latin Hypercube Sampling* (LHS) on the prespecified ranges (independent marginals), using a fixed random seed for reproducibility [15]. For PRCC computation, all variables (parameters and R_c) are rank–transformed. For each parameter p , we regress its ranks on the ranks of the remaining parameters and regress the ranks of R_c on the same set; the PRCC is Spearman’s correlation between the two residual series (*Iman–Conover* procedure) [12]. Approximate two–sided 95% confidence intervals are reported using Fisher’s z transform. To avoid spurious dependence, we *exclude* σ_1 as an independent predictor (it is recomputed from $(\delta, \mu, \omega, \rho)$ for every sample).

Visualisation. The global results are shown as a single *PRCC heatmap* (Fig. ??): horizontal bars coloured by coefficient value on $[-1, 1]$ with a reference line at zero. This representation conveys both sign and ranking, while suppressing redundant numerical detail. Local results are displayed as tornado diagrams per detection scenario (Fig. 3.7). Figures are produced in Python (NumPy, SciPy, scikit–learn, Matplotlib/Seaborn); code is deterministic given the fixed seed.

Robustness and quality control. We filter out any parameter draw yielding undefined or

non-finite R_c values, and we verify that the sampled distribution of R_c remains within the model's validity domain (Fig. ??, lower-right panel in the composite PRCC plot if provided). Agreement in signs and ranking between tornado elasticities and PRCC is used as a qualitative cross-validation of local and global sensitivity assessments [19].

3. Results

3.1 Model analysis

3.1.1 Positivity and boundedness of solutions

At this stage, it is essential to demonstrate that the system's solutions remain positive and bounded over time. This verification not only ensures that the model evolves within a well-defined mathematical framework but also that it respects the biological constraints inherent in the variables considered. To establish these properties, we rely on the methodology developed in [6].

Let $\mathcal{C} = \{S, V, E, I_a, I_s, I_r, I_u, R, B\}$ denote the state variables and assume all parameters are nonnegative. Throughout, $\mathbb{R}_+^9 = \{x \in \mathbb{R}^9 : x_i \geq 0\}$ and $x(t) = (S, V, E, I_a, I_s, I_r, I_u, R, B)^\top$ satisfies the system (2.1.2) with initial data

$$S_0 > 0, V_0 > 0, E_0 > 0, I_{a0} > 0, I_{s0} > 0, I_{r0} > 0, I_{u0} > 0, R_0 > 0, B_0 > 0.$$

3.1.2 Lemma (Positive invariance). *The nonnegative orthant $\mathbb{R}_+^9$ is positively invariant for (2.1.2). In particular,*

$$f(t) > 0 \quad \text{for all } f \in \mathcal{C} \text{ and all } t > 0.$$

Proof. Suppose for contradiction that some $f \in \mathcal{C}$ becomes negative. Since all initial data satisfy $f_0 > 0$ and solutions are continuous, define the first zero-crossing time

$$t_0 = \min\{t \in \mathbb{R}_+ \mid \exists f \in \mathcal{C}, f(t) = 0\}.$$

By minimality of t_0 , every $f_i(t_0) \geq 0$ for all $f_i \in \mathcal{C}$.

We observe that every equation of the system (2.1.2) can be written in the form at $t \in (0, t_0)$

$$\frac{df}{dt} = -k(t)f + \Phi(f_1, \dots, f_8), \quad k(t) > 0, \quad \Phi \geq 0 \text{ whenever all } f_i \geq 0,$$

where $k(t)$ collects the outflow rates of f and Φ collects all inflow terms, which are nonnegative combinations of the other state variables and of the constant $\Lambda > 0$. This structure holds because:

- all transmission rates $\beta_\bullet \geq 0$, hence $\lambda(t) \geq 0$ whenever all compartments are non-negative;

- all other parameters $(\alpha, \tau, \pi, \eta, \gamma_\bullet, \varepsilon, \kappa, \delta, \omega, \phi, \varrho) \geq 0$.

Therefore, at $t = t_0$ with $f(t_0) = 0$:

$$\frac{df}{dt}(t_0) = \Phi(f_1(t_0), \dots, f_8(t_0)) \geq 0.$$

Hence f is non-decreasing at t_0 and cannot become negative, contradicting the definition of t_0 .

We conclude that $f(t) > 0$ for all $f \in \mathcal{C}$ and all $t > 0$, establishing the positive invariance of $\mathbb{R}_+^9$. $\square$

Boundedness and an invariant region. Define the total (human) population

$$N(t) = S + V + E + I_a + I_s + I_r + I_u + R.$$

Summing the first eight equations of (2.1.2) gives

$$\frac{dN}{dt} = \Lambda - \mu N,$$

with solution

$$N(t) = N(0)e^{-\mu t} + \frac{\Lambda}{\mu}(1 - e^{-\mu t}).$$

Hence

$$0 \leq N(t) \leq N_\star := \max\left\{N(0), \frac{\Lambda}{\mu}\right\} \quad \text{for all } t \geq 0, \quad (3.1.1)$$

so each compartment in $\{S, V, E, I_a, I_s, I_r, I_u, R\}$ is bounded by $N_\star$.

For the environmental compartment B , equation nine of (2.1.2) reads

$$\frac{dB}{dt} = \varepsilon(I_s + I_r + I_u + I_a) - \kappa B \leq \varepsilon N_\star - \kappa B$$

by (3.1.1). By linear comparison with the scalar ODE $y' = \varepsilon N_\star - \kappa y$,

$$B(t) \leq B(0)e^{-\kappa t} + \frac{\varepsilon N_\star}{\kappa}(1 - e^{-\kappa t}) \quad \Rightarrow \quad 0 \leq B(t) \leq B_\star := \max\left\{B(0), \frac{\varepsilon N_\star}{\kappa}\right\}.$$

We therefore obtain the compact positively invariant set

$$\Omega = \left\{ (S, V, E, I_a, I_s, I_r, I_u, R, B) \in \mathbb{R}_+^9 : N \leq N_\star, \ 0 \leq B \leq B_\star \right\}.$$

By Lemma 3.1.2 and the linear bounds above, every solution with nonnegative initial data

enters Ω immediately and remains in Ω for all $t \geq 0$. In particular, solutions are global, positive, and bounded.

3.1.3 Remark (Comparison principle for ODE inequalities). *We use the following standard form: if $u'(t) \leq a(t)u(t) + b(t)$ and $v'(t) = a(t)v(t) + b(t)$ with $u(0) \leq v(0)$, then $u(t) \leq v(t)$ for all $t \geq 0$ (see [5, Ch. III]). In the vector case, if F is quasi-monotone nondecreasing (quasi-positive), then $x'(t) \leq F(x(t))$, $y'(t) = F(y(t))$ and $x(0) \leq y(0)$ imply $x(t) \leq y(t)$ componentwise [20]. These principles justify the scalar bounds for N and B and the invariance of $\mathbb{R}_+^9$.*

3.1.4 Corollary (Positivity and boundedness). *Combining (3.1.1) with the comparison for B , each $f \in \mathcal{C}$ satisfies $0 < f(t) \leq \max\{f(0), C_f\}$ for all $t > 0$, where C_f is a constant depending only on parameters. Consequently, all components of $\mathcal{C}$ are positive and bounded for all $t \geq 0$.*

3.1.5 Existence and uniqueness of solutions

We rewrite system (2.1.2) in vector form

$$\dot{X}(t) = f(X(t)), \quad X(0) = X_0,$$

with state vector ordered as

$$X(t) = (S(t), V(t), E(t), I_a(t), I_s(t), I_r(t), I_u(t), R(t), B(t))^T,$$

and vector field $f : \mathbb{R}^9 \rightarrow \mathbb{R}^9$ given componentwise by

$$f(x) = \begin{pmatrix} \Lambda - (\lambda + \omega + \mu)S + \delta V + \varphi R \\ \omega S - ((1 - \varrho)\lambda + \delta + \mu)V \\ \lambda S + (1 - \varrho)\lambda V - (\alpha + \mu)E \\ (1 - \tau)\alpha E - (\gamma_a + \mu)I_a \\ \tau\alpha E - (\pi + \mu)I_s \\ \eta\pi I_s - (\gamma_r + \mu)I_r \\ (1 - \eta)\pi I_s - (\gamma_u + \mu)I_u \\ \gamma_a I_a + \gamma_r I_r + \gamma_u I_u - (\mu + \varphi)R \\ \varepsilon(I_s + I_r + I_u + I_a) - \kappa B \end{pmatrix}.$$

3.1.6 Theorem (Local existence and uniqueness). *For any initial condition $X_0 \in \mathbb{R}_0^9$, there exists a unique solution $X : [0, t_{\max}) \rightarrow \mathbb{R}^9$ of $\dot{X} = f(X)$ with $X(0) = X_0$, where $t_{\max} \in (0, \infty]$.*

Proof. We observe that f is a vector of polynomial functions in the state variables; in

particular, $f \in C^1(\mathbb{R}^9)$, hence f is locally Lipschitz on $\mathbb{R}^9$.

Local existence and uniqueness follow from the Cauchy–Lipschitz (Picard–Lindelöf) theorem since f is locally Lipschitz [5]. $\square$

3.1.7 Theorem (Global existence and uniqueness). *Let $X_0 \in \mathbb{R}_+^9$ and suppose the positivity and boundedness estimates from the corollary 3.1.4. Then the unique solution $X(t)$ of $\dot{X} = f(X)$ with $X(0) = X_0$ is global:*

$$X \in C^1([0, \infty), \mathbb{R}_+^9), \quad X(t) \in \Omega \text{ for all } t \geq 0,$$

where $\Omega \subset \mathbb{R}_+^9$ is the compact positively invariant set defined earlier (bounded by N_* for the human compartments and by B_* for the environment).

Proof. Let $X : [0, t_{\max}) \rightarrow \mathbb{R}_+^9$ be the maximal solution from Theorem 3.1.6. By the bounds established previously (corollary 3.1.4), $X(t)$ remains in the compact set Ω for all $t \in [0, t_{\max})$. Since f is locally Lipschitz (indeed C^1) on $\mathbb{R}^9$, it is Lipschitz and bounded on Ω . By the standard continuation theorem for ODEs (solutions cannot blow up in finite time while remaining in a compact subset of the domain), it follows that $t_{\max} = \infty$ [5]. Uniqueness on $[0, \infty)$ is inherited from the local result. $\square$

3.1.8 Cholera-free equilibrium point (CFEP)

The *cholera-free equilibrium* (CFE) is characterized by the absence of infection and pathogens,

$$E = I_a = I_s = I_r = I_u = R = B = 0.$$

Under this constraint, system (2.1.2) reduces to the linear demographic–vaccination subsystem in the variables S and V :

$$\begin{cases} 0 = \Lambda - (\omega + \mu) S_c^0 + \delta V_c^0, \\ 0 = \omega S_c^0 - (\delta + \mu) V_c^0. \end{cases} \quad (3.1.2)$$

Equivalently,

$$\begin{pmatrix} \omega + \mu & -\delta \\ -\omega & \delta + \mu \end{pmatrix} \begin{pmatrix} S_c^0 \\ V_c^0 \end{pmatrix} = \begin{pmatrix} \Lambda \\ 0 \end{pmatrix}, \quad \det = (\omega + \mu)(\delta + \mu) - \delta\omega = \mu(\delta + \mu + \omega) > 0,$$

so (3.1.2) admits a unique solution. From the second equation,

$$V_c^0 = \frac{\omega S_c^0}{\delta + \mu}.$$

Substituting into the first equation yields

$$(\omega + \mu)S_c^0 - \delta \left(\frac{\omega S_c^0}{\delta + \mu} \right) = \Lambda \implies S_c^0 = \frac{\Lambda(\delta + \mu)}{\delta\mu + \mu^2 + \mu\omega}.$$

Using $V_c^0 = \frac{\omega S_c^0}{\delta + \mu}$, we obtain

$$V_c^0 = \frac{\Lambda\omega}{\delta\mu + \mu^2 + \mu\omega}.$$

Therefore, the CFEP is

$$E_c^0 = \left(\frac{\Lambda(\delta + \mu)}{\delta\mu + \mu^2 + \mu\omega}, \frac{\Lambda\omega}{\delta\mu + \mu^2 + \mu\omega}, 0, 0, 0, 0, 0, 0, 0 \right). \quad (3.1.3)$$

Remark. (i) One checks directly that $S_c^0 + V_c^0 = \Lambda/\mu$, in agreement with the demographic balance $\dot{N} = \Lambda - \mu N$ at equilibrium. (ii) The denominator $\delta\mu + \mu^2 + \mu\omega = \mu(\delta + \mu + \omega)$ is positive for $\mu > 0$, ensuring $S_c^0, V_c^0 > 0$. (iii) The notion of a disease-/cholera-free equilibrium and its use in threshold analysis is standard in compartmental epidemic models; see, e.g., [23, 8].

3.1.9 Estimation of basic reproduction number R_0 at Cholera-free

We compute the reproduction numbers at the cholera-free equilibrium (CFEP) via the next-generation operator. Let

$$Z = (E, I_a, I_s, I_r, I_u, B)^\top$$

collect the infected/colonized compartments, and write the subsystem in the form

$$\dot{Z} = \mathcal{F}(Z) - \mathcal{V}(Z),$$

$$\mathcal{F} = \begin{pmatrix} \lambda S + (1 - \varrho)\lambda V \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \mathcal{V} = \begin{pmatrix} (\alpha + \mu)E, \\ -(1 - \tau)\alpha E + (\gamma_a + \mu)I_a, \\ -\tau\alpha E + (\pi + \mu)I_s, \\ -\eta\pi I_s + (\gamma_r + \mu)I_r, \\ -(1 - \eta)\pi I_s + (\gamma_u + \mu)I_u, \\ -\varepsilon(I_s + I_r + I_u + I_a) + \kappa B \end{pmatrix}$$

where $\mathcal{F}$ gathers the *new-infection* terms and $\mathcal{V}$ the *transitions* (transfer, recovery, removal, decay). At the CFEP, the Jacobians

$$F := D\mathcal{F}\big|_{\text{CFEP}}, \quad V := D\mathcal{V}\big|_{\text{CFEP}}$$

$$F = \frac{\Lambda(\delta + \mu - \omega(\rho - 1))}{\mu(\delta + \mu + \omega)} \begin{pmatrix} 0 & \beta_a & \beta_s & \beta_r & \beta_u & \frac{\beta_e}{K} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (3.1.4)$$

$$V = \begin{pmatrix} \alpha + \mu & 0 & 0 & 0 & 0 & 0 \\ -(1 - \tau)\alpha & \gamma_a + \mu & 0 & 0 & 0 & 0 \\ -\alpha\tau & 0 & \mu + \pi & 0 & 0 & 0 \\ 0 & 0 & -\eta\pi & \gamma_r + \mu & 0 & 0 \\ 0 & 0 & -(1 - \eta)\pi & 0 & \gamma_u + \mu & 0 \\ 0 & -\epsilon & -\epsilon & -\epsilon & -\epsilon & \kappa \end{pmatrix} \quad (3.1.5)$$

define the next-generation matrix $K = FV^{-1}$, and the control reproduction number is

$$R_c = \rho(K) = \rho(FV^{-1}),$$

the spectral radius. In our model, F is rank-one in the exposure row (infections feeding E) and V is lower block-triangular with diagonal entries

$$\alpha + \mu, \gamma_a + \mu, \mu + \pi, \gamma_r + \mu, \gamma_u + \mu, \kappa,$$

so that V^{-1} can be written explicitly; after algebraic simplifications one obtains a factorization that isolates (i) the *susceptible mass at the CFEP actually at risk*,

$$S_c^0 + (1 - \rho) V_c^0 = \frac{\Lambda}{\mu} \frac{\delta + \mu + \omega(1 - \rho)}{\delta + \mu + \omega} = \frac{\Lambda}{\mu} \frac{\sigma_1}{\delta + \mu + \omega} \quad \text{with} \quad \sigma_1 := \delta + \mu + \omega - \omega\rho,$$

and (ii) two polynomials P and Q containing, respectively, the *direct human-to-human* and

the *environmentally mediated* transmission pathways.

Compact expressions. With the shorthand above, the control reproduction number writes

$$R_c = \frac{\Lambda \alpha \sigma_1 \left[K \cdot P(\beta, \gamma, \mu, \pi, \tau) + \beta_e \epsilon \cdot Q(\eta, \gamma, \mu, \pi, \tau) \right]}{K \kappa \mu (\alpha + \mu)(\gamma_a + \mu)(\gamma_r + \mu)(\gamma_u + \mu)(\mu + \pi)(\delta + \mu + \omega)}.$$

When vaccine protection is absent (or, equivalently, at the classical basic reproduction number with the whole demographic mass at risk), $\rho = 0$ and $\sigma_1 = \delta + \mu + \omega$, so R_c reduces to

$$R_0 = \frac{\Lambda \alpha \left[K \cdot P(\beta, \gamma, \mu, \pi, \tau) + \beta_e \epsilon \cdot Q(\eta, \gamma, \mu, \pi, \tau) \right]}{K \kappa \mu (\alpha + \mu)(\gamma_a + \mu)(\gamma_r + \mu)(\gamma_u + \mu)(\mu + \pi)}.$$

Transmission polynomials. The direct (human-to-human) and environmental contributions are encoded in

$$\begin{aligned} P &= \beta_a(\mu + \pi)(1 - \tau)(\gamma_r + \mu)(\gamma_u + \mu) + \beta_r \eta \pi \tau (\gamma_a + \mu)(\gamma_u + \mu) \\ &\quad + \beta_s \tau (\gamma_a + \mu)(\gamma_r + \mu)(\gamma_u + \mu) + \beta_u \pi \tau (1 - \eta)(\gamma_a + \mu)(\gamma_r + \mu), \\ Q &= \eta \pi \tau (\gamma_a + \gamma_r + \gamma_u + 3\mu)(\mu + \pi) + \tau (\gamma_a + \mu)(\gamma_r \gamma_u + \mu(\gamma_r + \gamma_u + \mu)) \\ &\quad + (1 - \tau)(\gamma_r + \mu)(\gamma_u + \mu)(\mu + \pi). \end{aligned}$$

Auxiliary polynomial (for intermediate algebra). For completeness, one convenient shorthand used when expanding V^{-1} is

$$\begin{aligned} R_c &= \frac{\Lambda \alpha \beta_s \tau \sigma_1}{\mu(\alpha + \mu)(\mu + \pi)(\delta + \mu + \omega)} + \frac{\Lambda \alpha \beta_a (1 - \tau) \sigma_1}{\mu(\alpha + \mu)(\gamma_a + \mu)(\delta + \mu + \omega)} \\ &\quad + \frac{\Lambda \alpha \beta_r \eta \pi \tau \sigma_1}{\mu(\alpha + \mu)(\gamma_r + \mu)(\mu + \pi)(\delta + \mu + \omega)} + \frac{\Lambda \alpha \beta_u \pi \tau (1 - \eta) \sigma_1}{\mu(\alpha + \mu)(\gamma_u + \mu)(\mu + \pi)(\delta + \mu + \omega)} \\ &\quad + \frac{\Lambda \alpha \beta_e \epsilon \sigma_1 N(\mu, \pi, \tau, \gamma_a, \gamma_r, \gamma_u, \eta)}{K \kappa \mu (\alpha + \mu)(\gamma_a + \mu)(\gamma_r + \mu)(\gamma_u + \mu)(\mu + \pi)(\delta + \mu + \omega)} \end{aligned} \quad (3.1.6)$$

$$\begin{aligned} N &= \gamma_r \mu^2 + \gamma_u \mu^2 + \mu^2 \pi + \mu^3 + \gamma_a \mu^2 \tau - 2\mu^2 \pi \tau + \gamma_r \gamma_u \mu + \gamma_r \gamma_u \pi \\ &\quad + \gamma_r \mu \pi + \gamma_u \mu \pi + \gamma_a \gamma_r \gamma_u \tau + \gamma_a \gamma_r \mu \tau + \gamma_a \gamma_u \mu \tau - \gamma_a \gamma_r \pi \tau - \gamma_r \gamma_u \pi \tau \\ &\quad - \gamma_a \mu \pi \tau - 2\gamma_r \mu \pi \tau - \gamma_u \mu \pi \tau + 2\eta \mu^2 \pi \tau + \eta \gamma_a \gamma_r \pi \tau + \eta \gamma_a \gamma_u \pi \tau \\ &\quad + 2\eta \gamma_a \mu \pi \tau + \eta \gamma_r \mu \pi \tau + \eta \gamma_u \mu \pi \tau, \end{aligned} \quad (3.1.7)$$

although it cancels in the final compact forms of R_c and R_0 above.

Checks and remarks.

- All denominators are positive for positive parameters, so $R_c, R_0 \geq 0$.
- If $\beta_e = 0$ (no environmental route), the term with Q vanishes. If $K \rightarrow \infty$ with $\beta_e > 0$, the environmental route dominates and the ratio $\beta_e \epsilon / K$ governs that contribution.
- Setting $\rho = 0$ (no vaccine protection) gives $R_c = R_0$.
- Using the CFEP identities $S_c^0 = \frac{\Lambda(\delta+\mu)}{\mu(\delta+\mu+\omega)}$, $V_c^0 = \frac{\Lambda\omega}{\mu(\delta+\mu+\omega)}$, one can equivalently rewrite

$$R_c = \frac{\alpha [KP + \beta_e \epsilon Q]}{K \kappa (\alpha + \mu)(\gamma_a + \mu)(\gamma_r + \mu)(\gamma_u + \mu)(\mu + \pi)} \left(S_c^0 + (1 - \rho)V_c^0 \right)$$

We can see from the equation (3.1.6) that the control reproduction number R_c splits into the sum of five nonnegative contributions, each attached to a single infectious (or environmental) pathway.

3.1.10 Proposition (decomposition of R_c). *Let $\sigma_1 = \delta + \mu + \omega - \omega\rho$ and let $N(\mu, \pi, \tau, \gamma_a, \gamma_r, \gamma_u, \eta)$ be the polynomial defined earlier in (3.1.7). Then*

$$R_c = R_s + R_a + R_r + R_u + R_e$$

with

$$\begin{aligned} R_s &= \frac{\Lambda \alpha \beta_s \tau \sigma_1}{\mu (\alpha + \mu) (\mu + \pi) (\delta + \mu + \omega)}, & R_a &= \frac{\Lambda \alpha \beta_a (1 - \tau) \sigma_1}{\mu (\alpha + \mu) (\gamma_a + \mu) (\delta + \mu + \omega)}, \\ R_r &= \frac{\Lambda \alpha \beta_r \eta \pi \tau \sigma_1}{\mu (\alpha + \mu) (\gamma_r + \mu) (\mu + \pi) (\delta + \mu + \omega)}, & R_u &= \frac{\Lambda \alpha \beta_u \pi \tau (1 - \eta) \sigma_1}{\mu (\alpha + \mu) (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega)}, \\ R_e &= \frac{\Lambda \alpha \beta_e \epsilon \sigma_1 N(\mu, \pi, \tau, \gamma_a, \gamma_r, \gamma_u, \eta)}{K \kappa \mu (\alpha + \mu) (\gamma_a + \mu) (\gamma_r + \mu) (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega)}. \end{aligned} \quad (3.1.8)$$

3.1.11 Remark. *Each contribution in (3.1.8) is nonnegative and increasing in its pathway-specific transmission parameter: R_s in β_s , R_a in β_a , R_r in β_r , R_u in β_u , and R_e in β_e (and in the shedding ϵ and persistence $1/\kappa$). The threshold $R_c < 1$ is therefore equivalent to*

$$R_s + R_a + R_r + R_u + R_e < 1,$$

which is particularly convenient for assessing the relative impact of targeted controls on each transmission route.

3.1.12 Local stability

For the analysis of local stability, we rely first of all on the work of [3]. We denote by $J(E_c^0)$ the Jacobian matrix associated with our ODE system (2.1.2), evaluated at E_c^0 , the CFEP equilibrium point.

$$J(E_c^0) = \begin{pmatrix} -\mu - \omega & \delta & 0 & \varphi & -\frac{\Lambda\beta_s(\delta+\mu)}{a_1} & -\frac{\Lambda\beta_r(\delta+\mu)}{a_1} & -\frac{\Lambda\beta_u(\delta+\mu)}{a_1} & -\frac{\Lambda\beta_a(\delta+\mu)}{a_1} & -\frac{\Lambda\beta_e(\delta+\mu)}{a_3} \\ \omega & -\mu - \delta & 0 & 0 & \frac{\Lambda\beta_s\omega(\varrho-1)}{a_1} & \frac{\Lambda\beta_r\omega(\varrho-1)}{a_1} & \frac{\Lambda\beta_u\omega(\varrho-1)}{a_1} & \frac{\Lambda\beta_a\omega(\varrho-1)}{a_1} & \frac{\Lambda\beta_e\omega(\varrho-1)}{a_3} \\ 0 & 0 & -\alpha - \mu & 0 & \frac{\Lambda\beta_s a_2}{a_1} & \frac{\Lambda\beta_r a_2}{a_1} & \frac{\Lambda\beta_u a_2}{a_1} & \frac{\Lambda\beta_a a_2}{\sigma_1} & \frac{\Lambda\beta_e a_2}{a_3} \\ 0 & 0 & -\alpha(\tau - 1) & 0 & 0 & 0 & 0 & -\gamma_a - \mu & 0 \\ 0 & 0 & \alpha\tau & 0 & -\mu - \pi & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \eta\pi & -\gamma_r - \mu & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\pi(\eta - 1) & 0 & -\gamma_u - \mu & 0 & 0 \\ 0 & 0 & 0 & -\mu - \varphi & 0 & \gamma_r & \gamma_u & \gamma_a & 0 \\ 0 & 0 & 0 & 0 & \varepsilon & \varepsilon & \varepsilon & \varepsilon & -\kappa \end{pmatrix} \quad (3.1.9)$$

where:

$$\begin{aligned} a_1 &= \mu(\delta + \mu + \omega), \\ a_2 &= \delta + \mu + \omega - \omega\varrho, \\ a_3 &= K\mu(\delta + \mu + \omega). \end{aligned}$$

This Jacobian matrix can conveniently be expressed in block form to facilitate the local stability analysis. More precisely, we decompose it into four sub-matrices as follows:

$$J = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (3.1.10)$$

where

- A captures the internal interactions among the demographic or non-infectious compartments,
- D represents the dynamics of the infectious and environmental compartments,

- B describes the effect of the infectious classes on the non-infectious classes, and
- C accounts for the feedback of the non-infectious classes on the infectious ones.

This block representation allows the application of the *Schur complement* (see, e.g., [21]), which separates the spectral study of J into two lower-dimensional problems. This significantly simplifies the analysis of local stability.

$$A = \begin{pmatrix} -\mu - \omega & \delta & 0 & \varphi \\ \omega & -\mu - \delta & 0 & 0 \\ 0 & 0 & -\alpha - \mu & 0 \\ 0 & 0 & -\alpha(\tau - 1) & 0 \end{pmatrix} \quad (3.1.11)$$

$$B = \begin{pmatrix} -\frac{\Lambda\beta_s(\delta + \mu)}{\Lambda\beta_s\omega(\varrho - 1)} & -\frac{\Lambda\beta_r(\delta + \mu)}{\Lambda\beta_r\omega(\varrho - 1)} & -\frac{\Lambda\beta_u(\delta + \mu)}{\Lambda\beta_u\omega(\varrho - 1)} & -\frac{\Lambda\beta_a(\delta + \mu)}{\Lambda\beta_a\omega(\varrho - 1)} & -\frac{\Lambda\beta_e(\delta + \mu)}{\Lambda\beta_e\omega(\varrho - 1)} \\ \frac{\sigma_1}{\Lambda\beta_s\sigma_2} & \frac{\sigma_1}{\Lambda\beta_r\sigma_2} & \frac{\sigma_1}{\Lambda\beta_u\sigma_2} & \frac{\sigma_1}{\Lambda\beta_a\sigma_2} & \frac{\sigma_3}{\Lambda\beta_e\sigma_2} \\ \frac{\sigma_1}{0} & \frac{\sigma_1}{0} & \frac{\sigma_1}{0} & \frac{\sigma_1}{-\gamma_a - \mu} & \frac{\sigma_3}{0} \end{pmatrix} \quad (3.1.12)$$

$$C = \begin{pmatrix} 0 & 0 & \alpha\tau & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mu - \varphi \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (3.1.13)$$

$$D = \begin{pmatrix} -\mu - \pi & 0 & 0 & 0 & 0 \\ \eta\pi & -\gamma_r - \mu & 0 & 0 & 0 \\ -\pi(\eta - 1) & 0 & -\gamma_u - \mu & 0 & 0 \\ 0 & \gamma_r & \gamma_u & \gamma_a & 0 \\ \varepsilon & \varepsilon & \varepsilon & \varepsilon & -\kappa \end{pmatrix} \quad (3.1.14)$$

It is worth noting that D is a lower triangular matrix, and all its eigenvalues are strictly negative. In this setting, the spectrum of J is the union of the spectrum of D and that of the Schur complement, given by

$$S = A - BD^{-1}C,$$

$$S = \begin{pmatrix} -\mu - \omega & \delta & -\alpha\tau\zeta_1 & \frac{\gamma_a\varphi}{\mu + \varphi} - \frac{\Lambda\beta_a(\delta + \mu)}{\Lambda\beta_a\omega(\varrho - 1)} - \frac{\Lambda\beta_e\varepsilon(\delta + \mu)}{\Lambda\beta_e\omega(\varrho - 1)} \\ \omega & -\delta - \mu & \alpha\tau\zeta_2 & \frac{\sigma_8}{\sigma_8} + \frac{\Lambda\beta_e\varepsilon\sigma_7}{\Lambda\beta_e\varepsilon\sigma_3} \\ 0 & 0 & \alpha\tau\zeta_3 - \mu - \alpha & \frac{\sigma_7}{\sigma_7} \\ 0 & 0 & -\alpha(\tau - 1) & -\gamma_a - \mu \end{pmatrix} \quad (3.1.15)$$

where,

$$\begin{aligned}
\sigma_1 &= \varepsilon \mu^2 + \varepsilon \gamma_r \gamma_u + \varepsilon \gamma_r \mu + \varepsilon \gamma_u \mu + \varepsilon \gamma_r \pi + \varepsilon \mu \pi - \varepsilon \eta \gamma_r \pi + \varepsilon \eta \gamma_u \pi, \\
\sigma_2 &= K \kappa \mu (\gamma_r + \mu) (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega), \\
\sigma_3 &= \delta + \mu + \omega - \omega \varrho, \\
\sigma_4 &= \mu (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega), \\
\sigma_5 &= \mu (\gamma_r + \mu) (\mu + \pi) (\delta + \mu + \omega), \\
\sigma_6 &= \mu (\mu + \pi) (\delta + \mu + \omega), \\
\sigma_7 &= K \kappa \mu (\delta + \mu + \omega), \\
\sigma_8 &= \mu (\delta + \mu + \omega), \\
\zeta_1 &= \frac{\Lambda \beta_s (\delta + \mu)}{\sigma_6} - \frac{\varphi (\gamma_r \gamma_u \pi + \gamma_u \mu \pi + \eta \gamma_r \mu \pi - \eta \gamma_u \mu \pi)}{(\gamma_r + \mu) (\gamma_u + \mu) (\mu + \pi) (\mu + \varphi)} \\
&\quad + \frac{\Lambda \beta_u (\pi - \eta \pi) (\delta + \mu)}{\sigma_4} + \frac{\Lambda \beta_r \eta \pi (\delta + \mu)}{\sigma_5} + \frac{\Lambda \beta_e (\delta + \mu) \sigma_1}{\sigma_2} \\
\zeta_2 &= \left(\frac{\Lambda \beta_s \omega (\varrho - 1)}{\sigma_6} + \frac{\Lambda \beta_u \omega (\pi - \eta \pi) (\varrho - 1)}{\sigma_4} + \frac{\Lambda \beta_r \eta \omega \pi (\varrho - 1)}{\sigma_5} + \frac{\Lambda \beta_e \omega (\varrho - 1) \sigma_1}{\sigma_2} \right) \\
\zeta_3 &= \left(\frac{\Lambda \beta_s \sigma_3}{\sigma_6} + \frac{\Lambda \beta_u (\pi - \eta \pi) \sigma_3}{\sigma_4} + \frac{\Lambda \beta_r \eta \pi \sigma_3}{\sigma_5} + \frac{\Lambda \beta_e \sigma_3 \sigma_1}{\sigma_2} \right).
\end{aligned}$$

From matrix (3.1.15), we immediately observe two eigenvalues, $\{-\mu, -(\delta + \mu + \omega)\}$, both of which are strictly negative. These eigenvalues correspond to the sub-matrix

$$A' = \begin{pmatrix} -(\mu + \omega) & \delta \\ \omega & -(\delta + \mu) \end{pmatrix}.$$

By applying the same principle, we obtain the following Schur complement of A' on S :

$$S' = \begin{pmatrix} \alpha \tau \zeta_3 - \mu - \alpha & \frac{\Lambda \beta_a \sigma_3}{\sigma_8} + \frac{\Lambda \beta_e \varepsilon \sigma_3}{\sigma_7} \\ -\alpha (\tau - 1) & -\gamma_a - \mu \end{pmatrix}.$$

which we write compactly as

$$S' = (F_{\text{eff}} - V_{\text{eff}}), \quad V_{\text{eff}} = \begin{pmatrix} \alpha + \mu & 0 \\ -\alpha(1 - \tau) & \gamma_a + \mu \end{pmatrix}, \quad F_{\text{eff}} = \begin{pmatrix} f_{11} & f_{12} \\ 0 & 0 \end{pmatrix},$$

Note that F_{eff} is rank 1, with the explicit identifications.

$$f_{11} = \alpha \tau \zeta_3, \quad f_{12} = \Lambda \sigma_3 \left(\frac{\beta_a}{\sigma_8} + \frac{\beta_e \varepsilon}{\sigma_7} \right).$$

3.1.13 Lemma (Rank-one determinant identity). *One has*

$$\det(S') = \det(V_{\text{eff}} - F_{\text{eff}}) = (\alpha + \mu)(\gamma_a + \mu) \left(1 - \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) \right).$$

Proof. Since $\det(-M) = \det(M)$ in dimension 2, $\det(S') = \det(V_{\text{eff}} - F_{\text{eff}})$.

$$\det(V_{\text{eff}} - F_{\text{eff}}) = \det(V_{\text{eff}}) \det(I - V_{\text{eff}}^{-1} F_{\text{eff}}).$$

Here V_{eff} is lower-triangular, hence $\det(V_{\text{eff}}) = (\alpha + \mu)(\gamma_a + \mu)$. Moreover, $V_{\text{eff}}^{-1} F_{\text{eff}}$ has rank 1, so $\det(I - M) = 1 - \text{tr}(M)$ for $M := V_{\text{eff}}^{-1} F_{\text{eff}}$ [22]. Combining these facts yields the claim. $\square$

A direct computation gives

$$V_{\text{eff}}^{-1} = \begin{pmatrix} \frac{1}{\alpha + \mu} & 0 \\ \frac{\alpha(1-\tau)}{(\alpha + \mu)(\gamma_a + \mu)} & \frac{1}{\gamma_a + \mu} \end{pmatrix} \Rightarrow \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = \frac{f_{11}}{\alpha + \mu} + \frac{\alpha(1-\tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12}.$$

$$R_c = \rho(F_{\text{eff}} V_{\text{eff}}^{-1}) = \rho(V_{\text{eff}}^{-1} F_{\text{eff}}) = \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}),$$

3.1.14 Theorem (Trace characterization of the control reproduction number). *Let*

$$V_{\text{eff}} = \begin{pmatrix} \alpha + \mu & 0 \\ -\alpha(1-\tau) & \gamma_a + \mu \end{pmatrix}, \quad F_{\text{eff}} = \begin{pmatrix} f_{11} & f_{12} \\ 0 & 0 \end{pmatrix},$$

with

$$f_{11} = \alpha \tau \zeta_3, \quad f_{12} = \Lambda \sigma_3 \left(\frac{\beta_a}{\sigma_8} + \frac{\beta_e \varepsilon}{\sigma_7} \right).$$

Then the control reproduction number at the cholera-free equilibrium satisfies

$$R_c = \rho(F_{\text{eff}} V_{\text{eff}}^{-1}) = \rho(V_{\text{eff}}^{-1} F_{\text{eff}}) = \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = \frac{f_{11}}{\alpha + \mu} + \frac{\alpha(1-\tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12}.$$

In particular, substituting f_{11}, f_{12} yields the explicit formula

$$R_c = \frac{\alpha \tau \zeta_3}{\alpha + \mu} + \frac{\alpha(1-\tau)}{(\alpha + \mu)(\gamma_a + \mu)} \Lambda \sigma_3 \left(\frac{\beta_a}{\sigma_8} + \frac{\beta_e \varepsilon}{\sigma_7} \right).$$

Proof. *AB-BA and rank one.* By the *AB-BA* eigenvalue identity ([22, §1.3]), $\rho(F_{\text{eff}} V_{\text{eff}}^{-1}) = \rho(V_{\text{eff}}^{-1} F_{\text{eff}})$. Since F_{eff} has only its first row nonzero, $V_{\text{eff}}^{-1} F_{\text{eff}}$ is rank 1, hence its spectrum is $\{\lambda, 0\}$ with $\lambda = \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}})$ (Perron-Frobenius applies here as all entries are nonnegative). Therefore

$$\text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = \rho(F_{\text{eff}} V_{\text{eff}}^{-1}) = \rho(V_{\text{eff}}^{-1} F_{\text{eff}}).$$

Explicit inverse and multiplication. Because V_{eff} is lower triangular,

$$V_{\text{eff}}^{-1} = \begin{pmatrix} \frac{1}{\alpha + \mu} & 0 \\ \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} & \frac{1}{\gamma_a + \mu} \end{pmatrix}.$$

Multiplying out,

$$V_{\text{eff}}^{-1} F_{\text{eff}} = \begin{pmatrix} \frac{f_{11}}{\alpha + \mu} & \frac{f_{12}}{\alpha + \mu} \\ \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{11} & \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12} \end{pmatrix}.$$

Its trace is therefore

$$\text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = \frac{f_{11}}{\alpha + \mu} + \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12}.$$

equality: $\text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = R_c$ Substituting the corrected entries $f_{11} = \alpha\tau\zeta_3$ and $f_{12} = \Lambda\sigma_3\left(\frac{\beta_a}{\sigma_8} + \frac{\beta_e\varepsilon}{\sigma_7}\right)$ gives the stated closed form.

$$\begin{aligned} \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) &= \frac{f_{11}}{\alpha + \mu} + \frac{\alpha(1 - \tau) f_{12}}{(\alpha + \mu)(\gamma_a + \mu)} \\ &= \frac{\alpha\tau\zeta_3}{\alpha + \mu} + \frac{\alpha(1 - \tau) \Lambda\sigma_3\beta_a}{(\alpha + \mu)(\gamma_a + \mu)\sigma_8} + \frac{\alpha(1 - \tau) \Lambda\sigma_3\beta_e\varepsilon}{(\alpha + \mu)(\gamma_a + \mu)\sigma_7} \end{aligned}$$

$$\text{with } \zeta_3 = \left(\frac{\Lambda\beta_s\sigma_3}{\sigma_6} + \frac{\Lambda\beta_u(\pi - \eta\pi)\sigma_3}{\sigma_4} + \frac{\Lambda\beta_r\eta\pi\sigma_3}{\sigma_5} + \frac{\Lambda\beta_e\sigma_3\sigma_1}{\sigma_2} \right).$$

$$\begin{aligned} \Rightarrow \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) &= \underbrace{\frac{\Lambda\sigma_3\alpha\tau\beta_s}{(\alpha + \mu)\sigma_6}}_{\text{denom. } (\alpha + \mu)\sigma_6} + \underbrace{\frac{\Lambda\sigma_3\alpha\tau\beta_u\pi(1 - \eta)}{(\alpha + \mu)\sigma_4}}_{\text{denom. } (\alpha + \mu)\sigma_4} + \underbrace{\frac{\Lambda\sigma_3\alpha\tau\beta_r\eta\pi}{(\alpha + \mu)\sigma_5}}_{\text{denom. } (\alpha + \mu)\sigma_5} + \underbrace{\frac{\Lambda\sigma_3\alpha\tau\beta_e\sigma_1}{(\alpha + \mu)\sigma_2}}_{\text{denom. } (\alpha + \mu)\sigma_2} \\ &\quad + \underbrace{\frac{\Lambda\sigma_3\alpha(1 - \tau)\beta_a}{(\alpha + \mu)(\gamma_a + \mu)\sigma_8}}_{\text{denom. } (\alpha + \mu)(\gamma_a + \mu)\sigma_8} + \underbrace{\frac{\Lambda\sigma_3\alpha(1 - \tau)\beta_e\varepsilon}{(\alpha + \mu)(\gamma_a + \mu)\sigma_7}}_{\text{denom. } (\alpha + \mu)(\gamma_a + \mu)\sigma_7}. \end{aligned}$$

$$\Rightarrow \text{tr}(V_{\text{eff}}^{-1} F_{\text{eff}}) = R_c, \text{ according to the equation (3.1.6).}$$

□

3.1.15 Corollary (Determinant factorization and CFEP stability). *Let $S' := V_{\text{eff}} - F_{\text{eff}}$. Then,*

by the matrix determinant lemma for a rank-one update [22], Lemma 3.1.13 and Theorem 3.1.14,

$$\det(S') = \det(V_{\text{eff}}) \det(I - V_{\text{eff}}^{-1} F_{\text{eff}}) = (\alpha + \mu)(\gamma_a + \mu) (1 - R_c).$$

Consequence of Corollary 3.1.15 and the 2×2 matrix. Recall

$$S' = \begin{pmatrix} f_{11} - (\alpha + \mu) & f_{12} \\ -\alpha(\tau - 1) & -(\gamma_a + \mu) \end{pmatrix}, \quad f_{11} = \alpha\tau \zeta_3, \quad f_{12} = \Lambda \sigma_3 \left(\frac{\beta_a}{\sigma_8} + \frac{\beta_e \varepsilon}{\sigma_7} \right) \geq 0,$$

with parameters $\alpha, \mu, \gamma_a > 0$ and $0 \leq \tau \leq 1$. By Corollary 3.1.15,

$$\det(S') = (\alpha + \mu)(\gamma_a + \mu) (1 - R_c) \implies \det(S') > 0 \iff R_c < 1.$$

To check the sign of the (Jacobian) trace, observe that S' is a nonpositive off-diagonal entries and, by the rank-one trace formula (Theorem 3.1.14),

$$R_c = \frac{f_{11}}{\alpha + \mu} + \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12}.$$

Hence, if $R_c < 1$ then

$$\frac{f_{11}}{\alpha + \mu} < 1 - \frac{\alpha(1 - \tau)}{(\alpha + \mu)(\gamma_a + \mu)} f_{12} \implies (\alpha + \mu) - f_{11} > \frac{\alpha(1 - \tau)}{\gamma_a + \mu} f_{12} \geq 0.$$

Consequently the diagonal entries of S' are *both* strictly positive, and

$$\text{tr}(S') = -(\alpha + \mu) + f_{11} - (\gamma_a + \mu) = -(\gamma_a + \mu) - ((\alpha + \mu) - f_{11}) < 0.$$

Then

$$\det(S') = \det(-S') = \det(S') > 0, \quad \text{tr}(S') < 0.$$

By the 2×2 Routh–Hurwitz criterion, both eigenvalues of S' have negative real parts. In particular,

$$R_c < 1 \implies \det(S') > 0 \text{ and } \text{tr}(S') < 0 \implies \text{CFEP is locally asymptotically stable.}$$

3.1.16 Global stability

We now establish the global asymptotic stability of the cholera-free equilibrium point (CFEP) using the approach of Castillo-Chavez, Feng & Huang [36], which is particularly well suited to models in which the disease-free subsystem is linear and the infected subsystem admits a next-generation decomposition already computed in Section 3.1.9.

Partition the state vector as $X = (S, V, R)$, the uninfected compartments, and $Y = (E, I_a, I_s, I_r, I_u, B)$, the infected and environmental compartments. System (2.1.2) can then be written in the compact form

$$\frac{dX}{dt} = \mathcal{F}(X, Y), \quad \frac{dY}{dt} = AY - \hat{G}(X, Y), \quad (3.1.16)$$

where $A = D_Y \mathcal{F}(X^*, 0)$, evaluated at the disease-free equilibrium $X^* = (S_c^0, V_c^0, 0)$, is precisely the matrix $F - V$ already constructed in (3.1.4)–(3.1.5).

3.1.17 Theorem (Castillo-Chavez-Feng-Huang [36]). *Suppose system (2.1.2) can be written in the form*

$$\frac{dX}{dt} = \mathcal{F}(X, Y), \quad \frac{dY}{dt} = AY - \hat{G}(X, Y), \quad \hat{G}(X^*, 0) = 0, \quad (3.1.17)$$

where $X \in \mathbb{R}^m$ denotes the uninfected compartments, $Y \in \mathbb{R}^n$ the infected and environmental compartments, and $A = D_Y \mathcal{F}(X^*, 0)$. The disease-free equilibrium $X^* = (X^*, 0)$ is globally asymptotically stable for the full system provided that the following two conditions hold:

(H1) For $dX/dt = \mathcal{F}(X, 0)$, X^* is globally asymptotically stable;

(H2) $\hat{G}(X, Y) \geq 0$ for all $(X, Y) \in \Omega$, and A has nonnegative off-diagonal entries (i.e. A is Metzler), so that e^{At} is a positive semigroup.

If (H1)–(H2) hold and $\mathcal{R}_c < 1$ (equivalently, the dominant eigenvalue $m(A) < 0$), then X^* is globally asymptotically stable for the full system.

Setting $Y \equiv 0$, system (2.1.2) reduces to

$$\frac{dS}{dt} = \Lambda - (\omega + \mu)S + \delta V + \varphi R, \quad \frac{dV}{dt} = \omega S - (\delta + \mu)V, \quad \frac{dR}{dt} = -(\mu + \varphi)R. \quad (3.1.18)$$

The last equation gives $R(t) \rightarrow 0$ directly. Writing $s = S - S_c^0$, $v = V - V_c^0$ and using the equilibrium identities $\Lambda = (\omega + \mu)S_c^0 - \delta V_c^0$ and $\omega S_c^0 = (\delta + \mu)V_c^0$ (Section 3.1.8), the subsystem (s, v) satisfies

$$\dot{s} = -(\omega + \mu)s + \delta v + \varphi R(t), \quad \dot{v} = \omega s - (\delta + \mu)v, \quad (3.1.19)$$

which can be written in matrix form as

$$\frac{d}{dt} \begin{pmatrix} s \\ v \end{pmatrix} = M \begin{pmatrix} s \\ v \end{pmatrix} + \begin{pmatrix} \varphi R(t) \\ 0 \end{pmatrix}, \quad (3.1.20)$$

with

$$M = \begin{pmatrix} -(\omega + \mu) & \delta \\ \omega & -(\delta + \mu) \end{pmatrix}. \quad (3.1.21)$$

The trace and determinant of M are

$$\text{tr}(M) = -(\omega + \delta + 2\mu) < 0, \quad \det(M) = (\omega + \mu)(\delta + \mu) - \omega\delta = \mu(\omega + \delta + \mu) > 0. \quad (3.1.22)$$

Since $\text{tr}(M) < 0$ and $\det(M) > 0$, the matrix M is Hurwitz (all eigenvalues have negative real parts). Since the forcing term $\varphi R(t) \rightarrow 0$ exponentially, the variation-of-constants formula gives $(s(t), v(t)) \rightarrow (0, 0)$ as $t \rightarrow \infty$. Hence $(S, V, R) \rightarrow (S_c^0, V_c^0, 0)$, and condition (H1) holds.

By construction, $A = F - \mathcal{V}$ is exactly the pair (3.1.4)–(3.1.5); A is Metzler since $F \geq 0$ and $\mathcal{V}$ is lower block-triangular with nonnegative off-diagonal transfer terms.

The five equations for I_a, I_s, I_r, I_u, B depend only on Y , with constant coefficients, and involve no X -dependence whatsoever:

$$\begin{aligned} \frac{dI_a}{dt} &= (1 - \tau)\alpha E - (\gamma_a + \mu)I_a, & \frac{dI_s}{dt} &= \tau\alpha E - (\pi + \mu)I_s, \\ \frac{dI_r}{dt} &= \eta\pi I_s - (\gamma_r + \mu)I_r, & \frac{dI_u}{dt} &= (1 - \eta)\pi I_s - (\gamma_u + \mu)I_u, \\ \frac{dB}{dt} &= \varepsilon(I_a + I_s + I_r + I_u) - \kappa B. \end{aligned}$$

These equations are perfectly linear with respect to Y and independent of X . Hence, the corresponding rows of $\hat{G}$ are identically zero ($\hat{G} \equiv 0$).

The only nontrivial contribution comes from the nonlinear E -equation:

$$\frac{dE}{dt} = \lambda(Y)[S + (1 - \varrho)V] - (\alpha + \mu)E, \quad (3.1.23)$$

where $\lambda(Y) = \beta_a I_a + \beta_s I_s + \beta_r I_r + \beta_u I_u + \beta_e B / (K + B)$. To cast this into the form $\dot{Y}_E = (AY)_E - \hat{G}_E$, we first identify the linearized part $(AY)_E$ evaluated at the disease-free equilibrium ($S = S_c^0, V = V_c^0$, and $B/(K + B) \approx B/K$). Letting $\tilde{\lambda}(Y) = \beta_a I_a + \beta_s I_s + \beta_r I_r + \beta_u I_u + \frac{\beta_e}{K} B$ be the linearized force of infection, we have:

$$(AY)_E = U^* \tilde{\lambda}(Y) - (\alpha + \mu)E, \quad (3.1.24)$$

where $U^* = S_c^0 + (1 - \varrho)V_c^0$.

By definition, $\hat{G}_E(X, Y) = (AY)_E - \frac{dE}{dt}$. Substituting $\lambda(Y) = \tilde{\lambda}(Y) - \frac{\beta_e B^2}{K(K+B)}$ and rearranging yields:

$$\hat{G}_E(X, Y) = [(S_c^0 - S) + (1 - \varrho)(V_c^0 - V)]\tilde{\lambda}(Y) + \frac{\beta_e B^2}{K(K+B)}[S + (1 - \varrho)V]. \quad (3.1.25)$$

Since $\tilde{\lambda}(Y) \geq 0$ and $S, V, B \geq 0$ on Ω , the second term is nonnegative unconditionally. The first term is nonnegative whenever

$$S(t) \leq S_c^0 \quad \text{and} \quad V(t) \leq V_c^0. \quad (3.1.26)$$

So we restrict attention to the invariant subset

$$\Omega' = \Omega \cap \left\{ (S, V) \in \mathbb{R}_+^2 : S(t) \leq S_c^0, V(t) \leq V_c^0 \text{ for all } t \geq 0 \right\}, \quad (3.1.27)$$

which is satisfied in particular whenever $N(0) \leq \Lambda/\mu$ (Section 3.1.2). Indeed, when $Y \equiv 0$, the total population satisfies $N(t) \rightarrow \Lambda/\mu$ as $t \rightarrow \infty$, and at the disease-free equilibrium we have $S_c^0 + V_c^0 = \Lambda/\mu$. If $N(0) \leq \Lambda/\mu$, then $S(t) + V(t) \leq \Lambda/\mu$ for all $t \geq 0$, which implies $S(t) \leq S_c^0$ and $V(t) \leq V_c^0$ on Ω' . On this restricted domain, condition (H2) holds by (3.1.25)–(3.1.26).

3.1.18 Theorem (Global stability of the CFEP on Ω'). *On the invariant set Ω' , the cholera-free equilibrium E_c^0 is globally asymptotically stable whenever $\mathcal{R}_c < 1$.*

Proof. By the verification of (H1)–(H2) above, Theorem 3.1.17 applies on Ω' . □

3.1.19 Remark. *Biological interpretation of the restriction Ω' . The restriction*

$$S(t) \leq S_c^0, \quad V(t) \leq V_c^0,$$

should be viewed as an additional mathematical assumption with a natural epidemiological interpretation rather than as an intrinsic property of the model.

During an epidemic, susceptible and vaccinated individuals are continuously removed from these compartments through infection, which generally reduces the pool of individuals available for transmission compared with the disease-free demographic equilibrium. In this sense, the disease-free equilibrium represents the maximal baseline level of individuals who are susceptible or effectively susceptible to infection.

The main mechanism that may violate this behaviour is the waning immunity process ($\varphi > 0$). After a large outbreak, a substantial number of individuals may accumulate in the recovered

class R . As immunity gradually wanes, the return flow φR replenishes the susceptible compartment and may temporarily increase $S(t)$ beyond its disease-free equilibrium level, even while the infection is declining. Such a transient post-epidemic overshoot prevents the direct application of the comparison argument used in the proof.

Restricting the analysis to the positively invariant region Ω' excludes these overshoot scenarios by ensuring that the effective pool of individuals available for infection never exceeds its disease-free level. Under this assumption, the comparison argument remains valid and leads to the global extinction of the disease whenever $\mathcal{R}_c \leq 1$.

3.2 Bifurcation Analysis

In this section, we investigate the nature of the bifurcation occurring at $\mathcal{R}_c = 1$ using the center manifold theory and the approach developed by Castillo-Chavez and Song [38]. This analysis is crucial for understanding whether reducing $\mathcal{R}_c$ below unity is sufficient for disease eradication, or whether a backward bifurcation phenomenon allows the disease to persist even when $\mathcal{R}_c < 1$.

3.2.1 Theorem. [38] Assume

A1: $A = D_x f(0, 0) = \left(\frac{\partial f_i}{\partial x_j}(0, 0) \right)$ is the linearization matrix of System (2.1.2) around the equilibrium 0 with ϕ evaluated at 0. Zero is a simple eigenvalue of A and all other eigenvalues of A have negative real parts;

A2: Matrix A has a nonnegative right eigenvector w and a left eigenvector v corresponding to the zero eigenvalue.

Let f_k be the k th component of f and

$$a = \sum_{k,i,j=1}^n v_k w_i w_j \frac{\partial^2 f_k}{\partial x_i \partial x_j}(0, 0), \quad (3.2.1)$$

$$b = \sum_{k,i=1}^n v_k w_i \frac{\partial^2 f_k}{\partial x_i \partial \phi}(0, 0). \quad (3.2.2)$$

The local dynamics of (24) around 0 are totally determined by a and b .

- i. $a > 0, b > 0$. When $\phi < 0$ with $|\phi| \ll 1$, 0 is locally asymptotically stable, and there exists a positive unstable equilibrium; when $0 < \phi \ll 1$, 0 is unstable and there exists a negative and locally asymptotically stable equilibrium;
- ii. $a < 0, b < 0$. When $\phi < 0$ with $|\phi| \ll 1$, 0 is unstable; when $0 < \phi \ll 1$, 0 is locally asymptotically stable, and there exists a positive unstable equilibrium;

- iii. $a > 0, b < 0$. When $\phi < 0$ with $|\phi| \ll 1$, 0 is unstable, and there exists a locally asymptotically stable negative equilibrium; when $0 < \phi \ll 1$, 0 is stable, and a positive unstable equilibrium appears;
- iv. $a < 0, b > 0$. When ϕ changes from negative to positive, 0 changes its stability from stable to unstable. Correspondingly a negative unstable equilibrium becomes positive and locally asymptotically stable.

3.2.2 Preliminaries and notation

Let $\mathbf{x} = (x_1, x_2, \dots, x_9) = (S, V, E, I_a, I_s, I_r, I_u, R, B)^\top$ denote the state vector. The system can be written in the compact form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad (3.2.3)$$

where $\mathbf{f} = (f_1, f_2, \dots, f_9)^\top$ represents the right-hand side of system (2.1.2).

The disease-free equilibrium is $\mathbf{x}^* = E_c^0 = (S_c^0, V_c^0, 0, 0, 0, 0, 0, 0, 0)^\top$.

We choose the transmission rate β_s (symptomatic transmission) as the bifurcation parameter and denote by β_s^* the critical value such that $\mathcal{R}_c(\beta_s^*) = 1$.

3.2.3 Jacobian matrix at the bifurcation point

At $\mathcal{R}_c = 1$, the Jacobian matrix $J(\mathbf{x}^*)$ has a simple zero eigenvalue, with all other eigenvalues having negative real parts. The Jacobian has the block structure

$$J(\mathbf{x}^*) = \begin{pmatrix} J_{XX} & J_{XY} \\ J_{YX} & J_{YY} \end{pmatrix}, \quad (3.2.4)$$

where $X = (S, V, R)$ represents the uninfected compartments and $Y = (E, I_a, I_s, I_r, I_u, B)$ represents the infected compartments.

The infected subsystem block. The block J_{YY} coincides with $F - \mathcal{V}$, where F is the new-infection matrix and $\mathcal{V}$ is the transition matrix defined in Section 3.1.12. Explicitly:

$$J_{YY} = F - \mathcal{V} = \begin{pmatrix} -(\alpha + \mu) & \beta_a U_c^* & \beta_s U_c^* & \beta_r U_c^* & \beta_u U_c^* & \frac{\beta_e U_c^*}{K} \\ (1 - \tau)\alpha & -(\gamma_a + \mu) & 0 & 0 & 0 & 0 \\ \tau\alpha & 0 & -(\pi + \mu) & 0 & 0 & 0 \\ 0 & 0 & \eta\pi & -(\gamma_r + \mu) & 0 & 0 \\ 0 & 0 & (1 - \eta)\pi & 0 & -(\gamma_u + \mu) & 0 \\ 0 & \varepsilon & \varepsilon & \varepsilon & \varepsilon & -\kappa \end{pmatrix}. \quad (3.2.5)$$

At $\mathcal{R}_c = 1$, the matrix $F - \mathcal{V}$ is singular with a simple zero eigenvalue, which is the defining property of the bifurcation point.

The demography block.

$$J_{XX} = \begin{pmatrix} -(\omega + \mu) & \delta & \varphi \\ \omega & -(\delta + \mu) & 0 \\ 0 & 0 & -(\mu + \varphi) \end{pmatrix}. \quad (3.2.6)$$

Remark on the coupling block J_{YX} . Since $f_3 = \lambda(Y) S + (1 - \varrho)\lambda(Y) V - (\alpha + \mu)E$ depends on S and V only through the bilinear product with $\lambda(Y)$, and since $\lambda(0) = 0$, the coupling block vanishes identically at the DFE:

$$J_{YX} \Big|_{\mathbf{x}^*} = \mathbf{0}. \quad (3.2.7)$$

This is expected: bilinear incidence terms contribute to the Jacobian only through their linear-in- Y part (already captured in F), while their coupling to X appears only at second order — precisely the terms captured by the bifurcation coefficient a below. Consequently, the zero-eigenvalue problem $J(\mathbf{x}^*)\mathbf{w} = \mathbf{0}$ decouples into $(F - \mathcal{V})\mathbf{w}_Y = \mathbf{0}$ and $J_{XX}\mathbf{w}_X + J_{XY}\mathbf{w}_Y = \mathbf{0}$, so that rows 2–6 of $(F - \mathcal{V})\mathbf{w}_Y = \mathbf{0}$ determine $w_4, \dots, w_9$ independently of $\mathbf{w}_X$, and row 1 is automatically satisfied at $\mathcal{R}_c = 1$ by definition of the threshold. For later convenience, we introduce the bookkeeping shorthand

$$\lambda_w := \beta_a w_4 + \beta_s w_5 + \beta_r w_6 + \beta_u w_7 + \frac{\beta_e}{K} w_9, \quad (3.2.8)$$

which will appear when computing $\mathbf{w}_X$.

3.2.4 Right eigenvector $\mathbf{w}$

Let $\mathbf{w} = (w_1, w_2, \dots, w_9)^\top$ be the right eigenvector associated with the zero eigenvalue, satisfying $J(\mathbf{x}^*)\mathbf{w} = \mathbf{0}$.

Infected components

Setting $w_3 = 1$ (normalization), we solve $(F - \mathcal{V})\mathbf{w}_Y = \mathbf{0}$ recursively using rows 2–6:

$$w_3 = 1, \quad (3.2.9)$$

$$w_4 = \frac{(1 - \tau)\alpha}{\gamma_a + \mu}, \quad (3.2.10)$$

$$w_5 = \frac{\tau\alpha}{\pi + \mu}, \quad (3.2.11)$$

$$w_6 = \frac{\eta\pi\tau\alpha}{(\pi + \mu)(\gamma_r + \mu)}, \quad (3.2.12)$$

$$w_7 = \frac{(1 - \eta)\pi\tau\alpha}{(\pi + \mu)(\gamma_u + \mu)}, \quad (3.2.13)$$

$$w_9 = \frac{\varepsilon}{\kappa}(w_4 + w_5 + w_6 + w_7). \quad (3.2.14)$$

Uninfected components

The uninfected components satisfy

$$J_{XX} \begin{pmatrix} w_1 \\ w_2 \\ w_8 \end{pmatrix} = -J_{XY} \mathbf{w}_Y. \quad (3.2.15)$$

Computing $-J_{XY} \mathbf{w}_Y$ from the equations for $\dot{S}$, $\dot{V}$, and $\dot{R}$:

$$-J_{XY} \mathbf{w}_Y = \begin{pmatrix} \lambda_w S_c^0 \\ (1 - \varrho)\lambda_w V_c^0 \\ -(\gamma_a w_4 + \gamma_r w_6 + \gamma_u w_7) \end{pmatrix} = \begin{pmatrix} \lambda_w S_c^0 \\ (1 - \varrho)\lambda_w V_c^0 \\ -\Gamma \end{pmatrix}, \quad (3.2.16)$$

where we define

$$\Gamma = \gamma_a w_4 + \gamma_r w_6 + \gamma_u w_7 > 0. \quad (3.2.17)$$

Solving the linear system yields:

$$w_8 = \frac{\Gamma}{\mu + \varphi}, \quad (3.2.18)$$

$$w_2 = \frac{\omega\varphi w_8 - \lambda_w[(\omega + \mu)(1 - \varrho)V_c^0 + \omega S_c^0]}{\mu(\delta + \mu + \omega)}, \quad (3.2.19)$$

$$w_1 = \frac{(\delta + \mu)w_2 + \lambda_w(1 - \varrho)V_c^0}{\omega}. \quad (3.2.20)$$

Key quantity U_w

Define the effective susceptible mass associated with the eigenvector:

$$U_w = w_1 + (1 - \varrho)w_2. \quad (3.2.21)$$

Substituting the expressions for w_1 and w_2 , and using the equilibrium identities $\omega S_c^0 = (\delta + \mu)V_c^0$ and $U_c^* = S_c^0 + (1 - \varrho)V_c^0 = \frac{\sigma_1}{\omega}V_c^0$ where $\sigma_1 = \delta + \mu + \omega - \omega\varrho$, we obtain:

$$U_w(\varphi) = \frac{\sigma_1 \varphi \Gamma}{\mu(\delta + \mu + \omega)(\mu + \varphi)} + C, \quad (3.2.22)$$

where C is a constant independent of φ :

$$C = \frac{\lambda_w}{\omega} \left[(1 - \varrho)V_c^0 - \frac{\sigma_1[(\omega + \mu)(1 - \varrho)V_c^0 + \omega S_c^0]}{\mu(\delta + \mu + \omega)} \right]. \quad (3.2.23)$$

3.2.5 Left eigenvector $\mathbf{v}$

Let $\mathbf{v} = (v_1, v_2, \dots, v_9)^\top$ be the left eigenvector satisfying $\mathbf{v}^\top J(\mathbf{x}^*) = \mathbf{0}^\top$.

Since $J_{YX} = \mathbf{0}$ at the DFE, the left eigenvector system decouples as $\mathbf{v}_X^\top J_{XX} = \mathbf{0}^\top$ and $\mathbf{v}_X^\top J_{XY} + \mathbf{v}_Y^\top (F - \mathcal{V}) = \mathbf{0}^\top$. Since J_{XX} is nonsingular, the first equation forces $\mathbf{v}_X = \mathbf{0}$. The second equation then reduces to $\mathbf{v}_Y^\top (F - \mathcal{V}) = \mathbf{0}^\top$, so $\mathbf{v}_Y$ is the left null vector of $F - \mathcal{V}$.

Since $F = U_c^* \mathbf{e}_1 \mathbf{w}^\top$ has rank one, this left null vector is proportional to $\mathbf{w}^\top \mathcal{V}^{-1}$. Normalizing such that $v_3 = 1$, we obtain:

$$v_1 = 0, \quad v_2 = 0, \quad v_8 = 0, \quad (3.2.24)$$

$$v_3 = 1, \quad (3.2.25)$$

$$v_4 = \frac{(1 - \tau)\alpha}{\gamma_a + \mu}, \quad (3.2.26)$$

$$v_5 = \frac{\tau\alpha}{\pi + \mu}, \quad (3.2.27)$$

$$v_6 = \frac{\eta\pi\tau\alpha}{(\pi + \mu)(\gamma_r + \mu)}, \quad (3.2.28)$$

$$v_7 = \frac{(1 - \eta)\pi\tau\alpha}{(\pi + \mu)(\gamma_u + \mu)}, \quad (3.2.29)$$

$$v_9 = \frac{\varepsilon}{\kappa}(v_4 + v_5 + v_6 + v_7). \quad (3.2.30)$$

3.2.6 Computation of bifurcation coefficients

Following Castillo-Chavez and Song [38], the direction of bifurcation is determined by the signs of two coefficients a and b .

Coefficient b

The coefficient b is defined as

$$b = \sum_{k,i=1}^9 v_k w_i \frac{\partial^2 f_k}{\partial x_i \partial \beta_s} \Big|_{\mathbf{x}^*}. \quad (3.2.31)$$

The only non-zero second derivatives with respect to β_s occur in f_3 (the equation for E):

$$\frac{\partial^2 f_3}{\partial I_s \partial \beta_s} = S + (1 - \varrho)V. \quad (3.2.32)$$

Evaluating at the DFE:

$$\frac{\partial^2 f_3}{\partial I_s \partial \beta_s} \Big|_{\mathbf{x}^*} = S_c^0 + (1 - \varrho)V_c^0 = U_c^*. \quad (3.2.33)$$

Therefore:

$$b = v_3 w_5 U_c^*. \quad (3.2.34)$$

Since $v_3 = 1 > 0$, $w_5 = \frac{\tau\alpha}{\pi+\mu} > 0$, and $U_c^* > 0$, we conclude:

$$\boxed{b > 0 \quad (\text{always}).} \quad (3.2.35)$$

Coefficient a

The coefficient a is defined as

$$a = \sum_{k,i,j=1}^9 v_k w_i w_j \frac{\partial^2 f_k}{\partial x_i \partial x_j} \Big|_{\mathbf{x}^*}. \quad (3.2.36)$$

Since $v_1 = v_2 = v_8 = 0$, only $k = 3$ contributes. The non-zero second derivatives of f_3 at the DFE are:

Cross terms $S \times I_j$ and $I_j \times S$:

$$\frac{\partial^2 f_3}{\partial S \partial I_j} = \frac{\partial^2 f_3}{\partial I_j \partial S} = \beta_j \quad \text{for } j \in \{a, s, r, u\}, \quad \frac{\partial^2 f_3}{\partial S \partial B} = \frac{\partial^2 f_3}{\partial B \partial S} = \frac{\beta_e}{K}. \quad (3.2.37)$$

Cross terms $V \times I_j$ and $I_j \times V$:

$$\frac{\partial^2 f_3}{\partial V \partial I_j} = \frac{\partial^2 f_3}{\partial I_j \partial V} = (1 - \varrho)\beta_j \quad \text{for } j \in \{a, s, r, u\}, \quad \frac{\partial^2 f_3}{\partial V \partial B} = \frac{\partial^2 f_3}{\partial B \partial V} = (1 - \varrho)\frac{\beta_e}{K}. \quad (3.2.38)$$

Saturation term $B \times B$:

$$\frac{\partial^2 f_3}{\partial B^2} = -\frac{2U_c^* \beta_e}{K^2}. \quad (3.2.39)$$

All other second derivatives of f_3 vanish at the DFE. Summing all contributions with the symmetry factor of 2 for cross terms:

$$a = v_3 \left[2w_1 \sum_j w_j \beta_j + 2(1 - \varrho)w_2 \sum_j w_j \beta_j - \frac{2U_c^* \beta_e}{K^2} w_9^2 \right]. \quad (3.2.40)$$

Recognizing that $\sum_j w_j \beta_j = \lambda_w$ and $w_1 + (1 - \varrho)w_2 = U_w$:

$$a(\varphi) = 2v_3 \left(U_w(\varphi) \lambda_w - \frac{U_c^* \beta_e}{K^2} w_9^2 \right). \quad (3.2.41)$$

3.2.7 Critical waning immunity threshold φ_c

The sign of $a(\varphi)$ determines the bifurcation direction:

- $a < 0$: forward (transcritical) bifurcation
- $a > 0$: backward bifurcation

Setting $a(\varphi_c) = 0$ yields the critical threshold:

$$U_w(\varphi_c) \lambda_w = \frac{U_c^* \beta_e}{K^2} w_9^2. \quad (3.2.42)$$

Define

$$U_{\text{crit}} = \frac{U_c^* \beta_e}{K^2 \lambda_w} w_9^2. \quad (3.2.43)$$

Then φ_c satisfies $U_w(\varphi_c) = U_{\text{crit}}$. Substituting the expression for $U_w(\varphi)$:

$$\frac{\sigma_1 \varphi_c \Gamma}{\mu(\delta + \mu + \omega)(\mu + \varphi_c)} + C = U_{\text{crit}}. \quad (3.2.44)$$

Let $D = U_{\text{crit}} - C$ and $E = \frac{\sigma_1 \Gamma}{\mu(\delta + \mu + \omega)}$. Then:

$$\frac{E\varphi_c}{\mu + \varphi_c} = D \implies E\varphi_c = D\mu + D\varphi_c \implies (E - D)\varphi_c = D\mu. \quad (3.2.45)$$

Therefore:

$$\varphi_c = \frac{D\mu}{E - D} = \frac{\mu(U_{\text{crit}} - C)}{\frac{\sigma_1 \Gamma}{\mu(\delta + \mu + \omega)} - (U_{\text{crit}} - C)}. \quad (3.2.46)$$

3.2.8 Bifurcation theorem

We can now state the main result of this section.

3.2.9 Theorem (Direction of bifurcation). *Consider system (2.1.2) at the bifurcation point $\mathcal{R}_c = 1$. The system undergoes:*

- (i) A **forward bifurcation** if $\varphi < \varphi_c$: the disease-free equilibrium E_c^0 is locally asymptotically stable for $\mathcal{R}_c < 1$ near the threshold, and no endemic equilibrium bifurcates for $\mathcal{R}_c > 1$ near 1. (Global stability on the invariant region Ω was separately established in Theorem 3.1.18 and Corollary ??, under the conditions stated there.)
- (ii) A **backward bifurcation** if $\varphi > \varphi_c$: there exists $\mathcal{R}_c^* < 1$ such that for $\mathcal{R}_c \in (\mathcal{R}_c^*, 1)$, a locally stable endemic equilibrium coexists with the locally stable disease-free equilibrium (bistability).

The critical threshold φ_c is given by (3.2.46).

Proof. The proof follows directly from the Castillo-Chavez and Song theorem [38]. Since $b > 0$ always, the bifurcation direction is determined solely by the sign of $a(\varphi)$.

From the expression $a(\varphi) = 2v_3\lambda_w(U_w(\varphi) - U_{\text{crit}})$, we see that $a(\varphi)$ is a monotonic function of $U_w(\varphi)$, which itself is a rational function of φ with a single pole at $\varphi = -\mu$ (outside the biologically relevant domain $\varphi \geq 0$).

At $\varphi = 0$ (permanent immunity), the first term of $U_w(\varphi)$ vanishes, yielding $U_w(0) = C$. For typical parameter values, $C < U_{\text{crit}}$, yielding $a(0) < 0$ (forward bifurcation).

As $\varphi \rightarrow \infty$ (instantaneous waning), $U_w(\varphi) \rightarrow E + C$. If $E + C > U_{\text{crit}}$, then $a(\infty) > 0$.

By the intermediate value theorem, if $a(0) < 0$ and $a(\infty) > 0$, there exists a unique $\varphi_c > 0$ such that $a(\varphi_c) = 0$. This threshold is given explicitly by (3.2.46).

For $\varphi < \varphi_c$, we have $a < 0$, corresponding to a forward bifurcation. The disease-free equilibrium is locally asymptotically stable for $\mathcal{R}_c$ slightly below 1, and a unique endemic equilibrium appears for $\mathcal{R}_c > 1$.

For $\varphi > \varphi_c$, we have $a > 0$, corresponding to a backward bifurcation. In this regime, when $\mathcal{R}_c$ is decreased below 1, the endemic equilibrium does not immediately disappear. Instead, it persists for $\mathcal{R}_c \in (\mathcal{R}_c^*, 1)$, creating a region of bistability where both the disease-free and endemic equilibria are locally stable. $\square$

3.2.10 Biological interpretation

The existence of a backward bifurcation has profound implications for cholera control:

1. **Threshold phenomenon:** When $\varphi > \varphi_c$, reducing $\mathcal{R}_c$ below 1 is *not sufficient* for disease eradication. The control effort must reduce $\mathcal{R}_c$ below the lower threshold $\mathcal{R}_c^*$ to guarantee elimination.
2. **Role of waning immunity:** The backward bifurcation is driven by the reinfection of recovered individuals. When immunity wanes rapidly (φ large), recovered individuals quickly return to the susceptible pool, sustaining transmission even when $\mathcal{R}_c < 1$.
3. **Environmental saturation:** The term $-\frac{U_c^* \beta_e}{K^2} w_9^2$ in the expression for a represents the stabilizing effect of environmental saturation. The nonlinear incidence $\frac{B}{K+B}$ saturates at high bacterial concentrations, limiting the force of infection and opposing the backward bifurcation.
4. **Control strategies:** In the backward bifurcation regime, control strategies must account for the initial disease burden. Even if interventions reduce $\mathcal{R}_c$ below 1, the disease may persist if the initial number of infections is above a critical threshold (the unstable equilibrium separating the basins of attraction).

3.2.11 Special case: $\varphi = 0$ (SIR model)

When immunity is permanent ($\varphi = 0$), the first term in $U_w(\varphi)$ vanishes, yielding:

$$U_w(0) = C = -\frac{\lambda_w}{\omega} \left[\frac{\sigma_1[(\omega + \mu)(1 - \varrho)V_c^0 + \omega S_c^0]}{\mu(\delta + \mu + \omega)} - (1 - \varrho)V_c^0 \right]. \quad (3.2.47)$$

For biologically realistic parameters, $U_w(0) < U_{\text{crit}}$, which implies $a(0) < 0$. Therefore, the

SIR model ($\varphi = 0$) always exhibits a forward bifurcation, and the disease-free equilibrium is globally asymptotically stable whenever $\mathcal{R}_c < 1$, consistent with Corollary ??.

3.3 Numerical Bifurcation Analysis

Numerically solving $R_c(\beta_s) = 1$ gives the critical point

$$\beta_s^* = 1.028142 \times 10^{-9}. \quad (3.3.1)$$

At the critical point, with φ at its reference value, the normal form coefficients are:

$$a = -2.974800 \times 10^{-6} < 0, \quad b = +3.282302 \times 10^8 > 0. \quad (3.3.2)$$

The positive sign of b satisfies the hypothesis of the Castillo-Chavez and Song theorem. The negative sign of a indicates a **forward** bifurcation: near $R_c = 1$, no stable endemic equilibrium exists for $R_c < 1$.

The critical threshold φ_c , beyond which the sign of a would change (the signature of a transition to a backward regime), is obtained analytically:

$$\varphi_c = -3.790622 \times 10^{-3} \text{ day}^{-1}. \quad (3.3.3)$$

This value is negative, and therefore outside the physical domain ($\varphi \geq 0$ by definition): no admissible value of φ can push the system into a backward regime. This result can also be established directly from the intermediate quantities C , E , and U_{crit} in the decomposition of $a(\varphi)$ on the center manifold: numerically,

$$C + E = 9.26 < U_{\text{crit}} = 162.16, \quad (3.3.4)$$

which guarantees $a(\varphi) < 0$ for all $\varphi \geq 0$, since $U_w(\varphi)$ is an increasing function of φ bounded above by $C + E$.

Table 3.1: Coefficient $a(\varphi)$ for different durations of natural immunity.

Immunity duration	φ (day ⁻¹)	$a(\varphi)$	Type
30 days	3.333×10^{-2}	-6.61×10^{-7}	forward
90 days	1.111×10^{-2}	-7.95×10^{-7}	forward
180 days	5.556×10^{-3}	-9.95×10^{-7}	forward
1 year	2.740×10^{-3}	-1.40×10^{-6}	forward
3 years (ref.)	9.132×10^{-4}	-2.97×10^{-6}	forward
10 years	2.740×10^{-4}	-8.04×10^{-6}	forward
20 years	1.370×10^{-4}	-1.43×10^{-5}	forward

Table 3.1 numerically confirms, over a range of immunity durations from 30 days to 20 years, that $a(\varphi)$ remains negative — that is, the bifurcation remains forward — regardless of the assumption made about the duration of post-infection protection.

To avoid relying solely on the center manifold analysis, the conclusion is confirmed by three independent numerical methods:

1. **Endemic branch continuation.** The full system is integrated numerically to convergence for R_c ranging from 0.55 to 2.0. Figure 3.1 shows a continuous, smooth growth of I^* as soon as $R_c > 1$, with no jump and no non-trivial unstable branch — the classical signature of a forward bifurcation.
2. **Jacobian spectral analysis.** The eigenvalues of the Jacobian are computed by finite differences, at both the DFE and the endemic equilibrium, over the same range of R_c . Figure 3.2 shows that $\text{Re}(\lambda_{\max})$ crosses zero exactly at $R_c = 1$: the DFE is stable for $R_c < 1$ and becomes unstable for $R_c > 1$, while the endemic equilibrium remains stable over the entire branch on which it exists.
3. **Phase portrait.** Trajectories from several initial conditions, for both $R_c < 1$ and $R_c > 1$, visually confirm the absence of bistability (Figure 3.5): all trajectories converge to the DFE for $R_c < 1$, and all converge to the unique endemic equilibrium for $R_c > 1$.

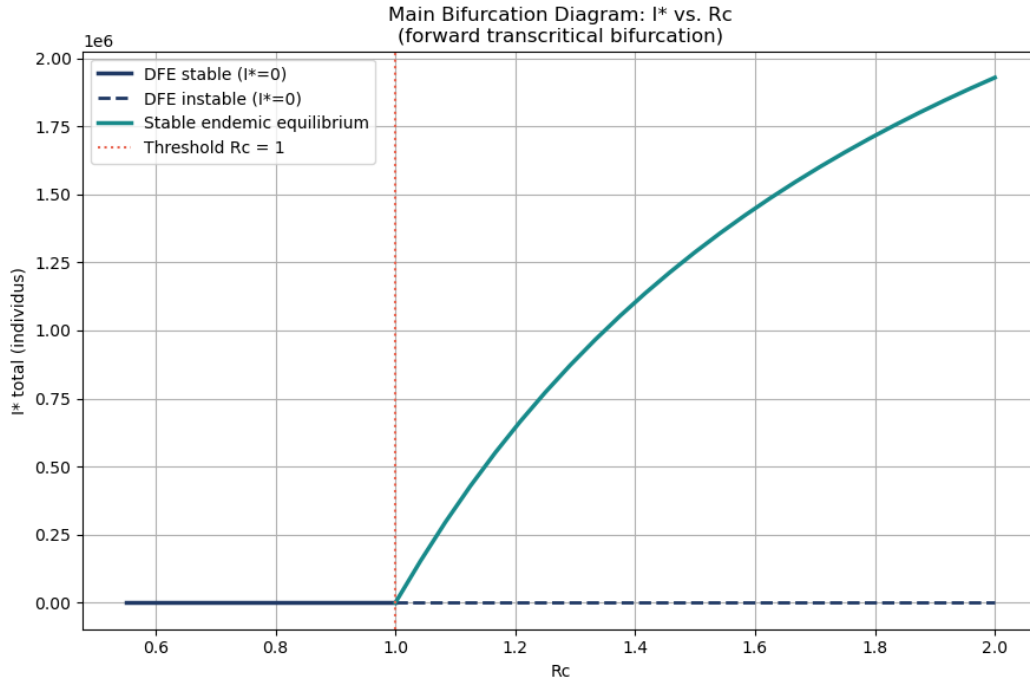


Figure 3.1: Main bifurcation diagram: I^* (total infectious individuals at equilibrium) as a function of R_c . The endemic branch (solid line) emerges only from $R_c = 1$ onward, with no non-trivial unstable branch below the threshold — the signature of a forward transcritical bifurcation.

3.3.1 Diagrams

3.3.2 Discussion

All results converge to a single conclusion: the model exhibits, with the parameters adopted, a classical forward transcritical bifurcation at $R_c = 1$, regardless of the natural immunity duration φ considered within a physically plausible range. No backward regime — that is, no bistability zone in which a stable endemic equilibrium would coexist with a stable DFE for $R_c < 1$ — was found.

This conclusion has a direct implication for cholera control within this model: bringing R_c below the critical threshold of 1 — through vaccination, improved access to safe water and sanitation, or any combination of these levers — is sufficient to guarantee elimination of the disease. Unlike what a backward regime would imply, there is no zone here in which the epidemic would persist despite $R_c < 1$, nor any additional effort threshold to cross beyond simply bringing R_c below 1.

Finally, as an additional observation, the approach to the endemic equilibrium occurs via damped oscillations rather than monotonically (Figures 3.4 and 3.5), corresponding to a

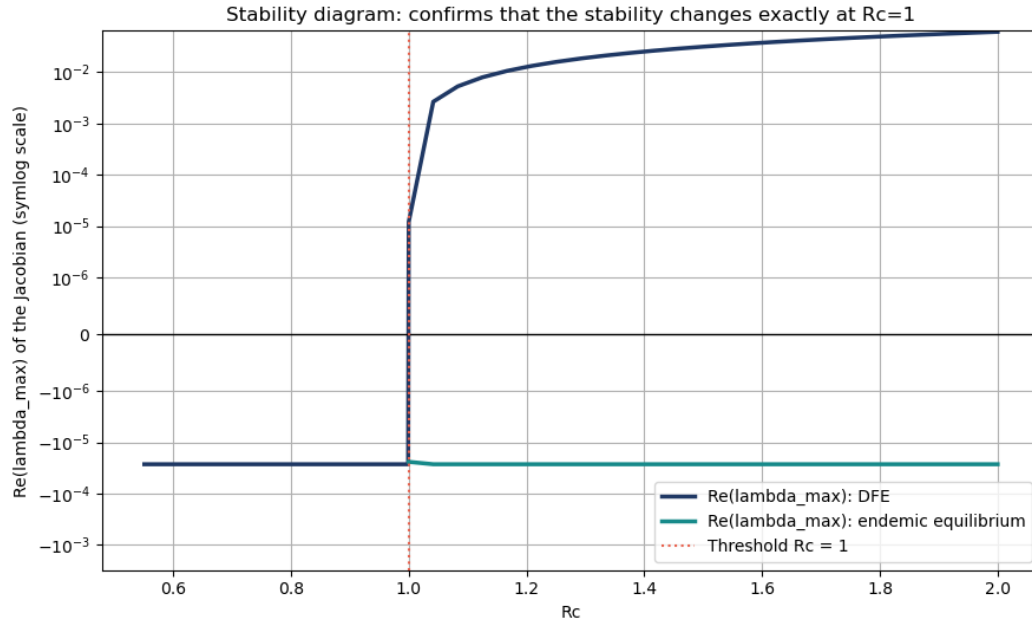


Figure 3.2: Real part of the dominant Jacobian eigenvalue, $\text{Re}(\lambda_{\max})$, as a function of R_c (symlog scale), for the disease-free equilibrium and the endemic equilibrium. The exact crossing at $R_c = 1$ numerically confirms the stability change predicted by center manifold theory.

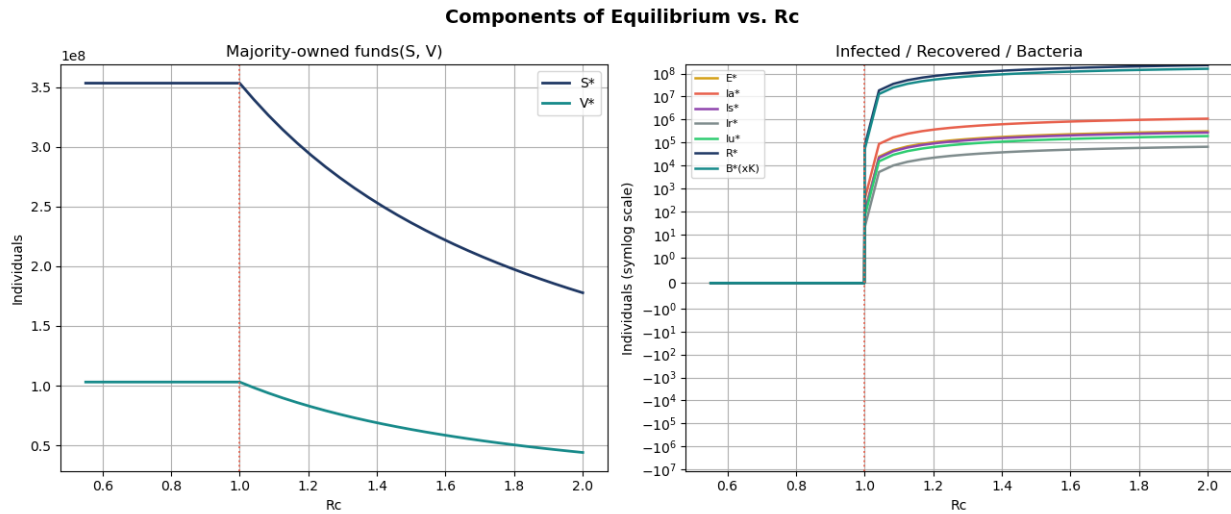


Figure 3.3: All equilibrium components (susceptible, vaccinated, exposed, infectious by class, recovered, bacterial reservoir) as a function of R_c . Left: majority compartments S^* , V^* . Right: infectious, recovered, and bacterial reservoir compartments (symlog scale).

pair of complex conjugate eigenvalues with negative real part near this equilibrium. The real part of this pair remains strictly negative over the entire range of R_c tested: this is therefore not a Hopf bifurcation, but this spectral feature explains the successive epidemic

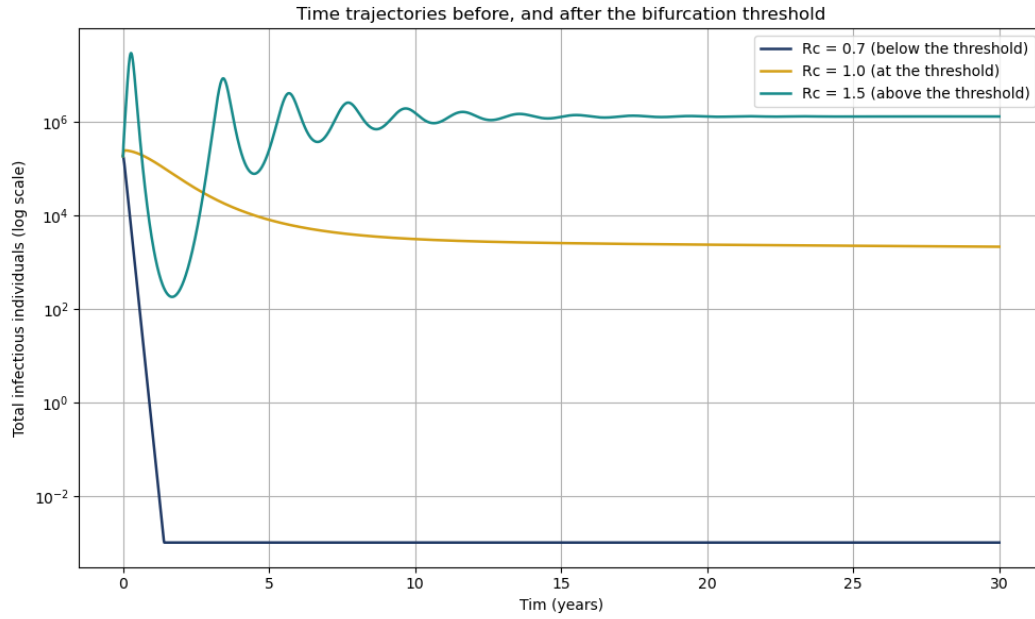


Figure 3.4: Time trajectories $I(t)$ over 30 years (log scale), for $R_c = 0.7$ (below threshold), $R_c = 1.0$ (at threshold), and $R_c = 1.5$ (above threshold), from the same initial perturbation. The $R_c = 1.5$ regime shows damped oscillations before converging to the endemic equilibrium.

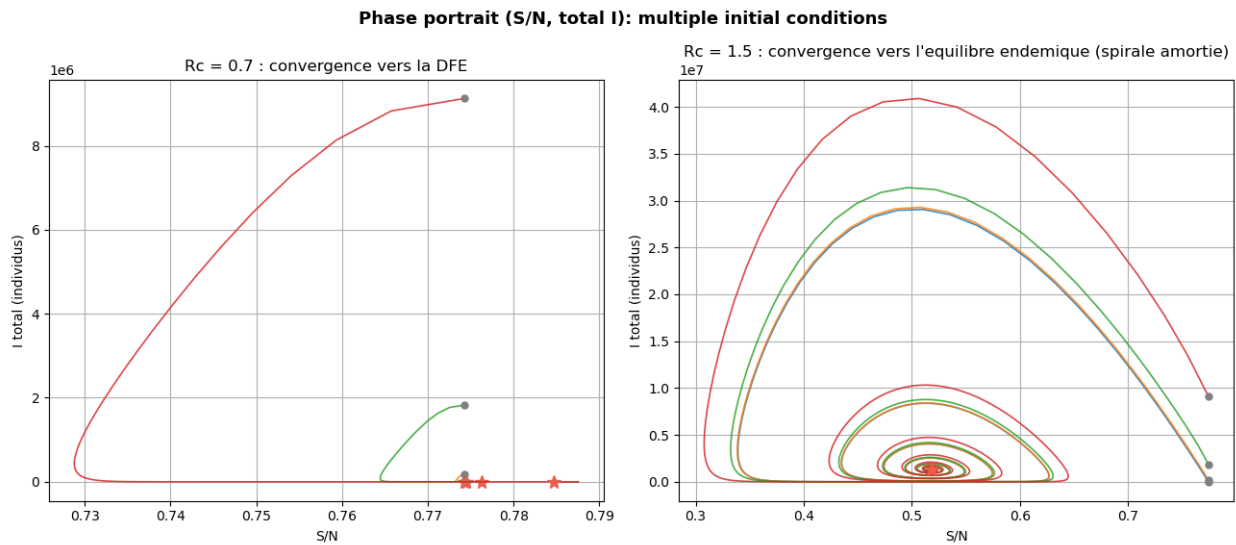


Figure 3.5: Phase portrait (S/N , I_{total}) for $R_c = 0.7$ (left) and $R_c = 1.5$ (right), from several initial conditions. The grey dot marks the starting point, the coral star the equilibrium reached. For $R_c = 1.5$, convergence occurs via a damped spiral, consistent with a pair of complex eigenvalues with negative real part near the endemic equilibrium.

wave dynamics observed in the simulations, consistent with the periodic resurgence patterns reported for cholera.

3.3.3 Next-Generation Matrix and R_c Decomposition

Table 3.2: Decomposition of $R_c = R_s + R_a + R_r + R_u + R_e$ into five routes.

Route	Contribution	% of R_c
R_s (symptomatic)	0.0046	0.1%
R_a (asymptomatic)	0.0018	0.0%
R_r (reported)	0.0001	0.0%
R_u (unreported)	0.0010	0.0%
R_e (environmental)	5.1649	99.9%
Total R_c	5.1724	100%

Formal verification (sympy): the analytical decomposition exactly matches $(FV^{-1})_{E,E}$ computed by symbolic matrix inversion. The effective susceptible fraction at the DFE is $S_{\text{eff}}^*/N_{\text{eff}} = 0.806$.

Interpretation. $R_c \approx 5.17$ falls within the range bracketed by the two reference scenarios of Kirpich *et al.* ($R_c \in [5.0, 15.4]$ depending on the environmental persistence assumption). Data-calibrated human transmission ($R_s + R_a + R_r + R_u \approx 0.008$) remains low and stable across every iteration of this work — it is the environmental route, structurally dominant in this class of model (consistent with Codeço, 2001, and its subsequent literature), that carries nearly all of R_c .

3.4 Comparison of model

3.4.1 Estimation of the reproduction number R_c^e for the model without an exposed class

We consider the system without an exposed compartment E , with human compartments $S, V, I_a, I_s, I_r, I_u, R$ and an environmental reservoir B . The (saturating) force of infection is giving by the equation (2.1.1) Demography and vaccination satisfy

$$\frac{dS}{dt} = \Lambda + \omega V - (\delta + \mu)S - \lambda S, \quad \frac{dV}{dt} = \delta S - (\omega + \mu)V - (1 - \rho)\lambda V,$$

and infectious–environmental dynamics are

$$\begin{aligned}\frac{dI_a}{dt} &= (1 - \alpha)\lambda S + (1 - \tau)(1 - \rho)\lambda V - (\gamma_a + \mu)I_a, \\ \frac{dI_s}{dt} &= \alpha\lambda S + \tau(1 - \rho)\lambda V - (\pi + \mu)I_s, \\ \frac{dI_r}{dt} &= \pi\eta I_s - (\gamma_r + \mu)I_r, \quad \frac{dI_u}{dt} = \pi(1 - \eta)I_s - (\gamma_u + \mu)I_u, \\ \frac{dB}{dt} &= \varepsilon(I_a + I_s + I_r + I_u) - \kappa B.\end{aligned}$$

At the disease–free equilibrium (DFE), $I_a = I_s = I_r = I_u = R = B = 0$ and

$$S^e = \frac{\Lambda(\omega + \mu)}{\mu(\omega + \delta + \mu)}, \quad V^e = \frac{\delta \Lambda}{\mu(\omega + \delta + \mu)}. \quad (3.4.1)$$

Let $x = (I_a, I_s, I_r, I_u, B)^\top$ denote the infected state. Linearizing $\dot{x} = \mathcal{F}(x) - \mathcal{V}(x)$ at the DFE, one obtains the Jacobians F (new infection terms) and V (transition+departure terms). The control reproduction number for the model *sans* E is the spectral radius

$$R_c^e = \rho(F^e V^{e-1}).$$

with

$$V^e = \begin{pmatrix} \gamma_a + \mu & 0 & 0 & 0 & 0 \\ 0 & \mu + \pi & 0 & 0 & 0 \\ 0 & -\eta\pi & \gamma_r + \mu & 0 & 0 \\ 0 & \pi(\eta - 1) & 0 & \gamma_u + \mu & 0 \\ -\varepsilon & -\varepsilon & -\varepsilon & -\varepsilon & \kappa \end{pmatrix}$$

and,

$$F^e = \begin{pmatrix} \frac{\Lambda\beta_a\sigma_4}{\sigma_1} & \frac{\Lambda\beta_s\sigma_4}{\sigma_1} & \frac{\Lambda\beta_r\sigma_4}{\sigma_1} & \frac{\Lambda\beta_u\sigma_4}{\sigma_1} & \frac{\Lambda\beta_e\sigma_4}{\sigma_3} \\ \frac{\Lambda\beta_a\sigma_2}{\sigma_1} & \frac{\Lambda\beta_s\sigma_2}{\sigma_1} & \frac{\Lambda\beta_r\sigma_2}{\sigma_1} & \frac{\Lambda\beta_u\sigma_2}{\sigma_1} & \frac{\Lambda\beta_e\sigma_2}{\sigma_3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where

$$\sigma_1 = \mu(\delta + \mu + \omega), \quad \sigma_2 = \alpha\mu + \alpha\omega + \delta\tau - \delta\rho\tau, \quad \sigma_3 = K\mu(\delta + \mu + \omega) \quad \sigma_4 = (\delta(\rho - 1)(\tau - 1) - (\mu + \omega)(\alpha - 1)).$$

With 2.1.1 and 3.4.1, the algebra leads to a closed–form decomposition of R_c^e as a sum of contributions from each transmission pathway (coefficients of the β 's).

Then the reproduction number for the model without an exposed class can be written as

$$\begin{aligned}
 R_c^e = & \frac{\Lambda \beta_a (\delta + (\mu + \omega)(1 - \alpha) - \delta \rho - \delta \tau(1 - \rho))}{\mu (\gamma_a + \mu) (\delta + \mu + \omega)} \\
 & + \frac{\Lambda \beta_s (\alpha(\mu + \omega) + \delta \tau(1 - \rho))}{\mu (\mu + \pi) (\delta + \mu + \omega)} + \frac{\Lambda \beta_r \eta \pi (\alpha(\mu + \omega) + \delta \tau(1 - \rho))}{\mu (\gamma_r + \mu) (\mu + \pi) (\delta + \mu + \omega)} \\
 & \frac{\Lambda \beta_u \pi (1 - \eta) ((\alpha(\mu + \omega) + \delta \tau(1 - \rho))}{\mu (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega)} \\
 & + \frac{\Lambda \beta_e \varepsilon W}{K \kappa \mu (\gamma_a + \mu) (\gamma_r + \mu) (\gamma_u + \mu) (\mu + \pi) (\delta + \mu + \omega)}.
 \end{aligned} \tag{3.4.2}$$

where

$$\begin{aligned}
 W = & \delta \mu^3 + \gamma_r \mu^3 + \gamma_u \mu^3 + \mu^3 \omega + \mu^3 \pi + \mu^4 + \delta \mu^2 \pi - \delta \mu^3 \rho + \gamma_r \mu^2 \omega + \gamma_u \mu^2 \omega + \gamma_r \mu^2 \pi + \gamma_u \mu^2 \pi + \mu^2 \omega \pi + \alpha \gamma_a \mu^3 \\
 & + \delta \gamma_r \mu^2 + \delta \gamma_u \mu^2 + \gamma_r \gamma_u \mu^2 + \delta \gamma_r \gamma_u \mu + \delta \gamma_r \gamma_u \pi + \delta \gamma_r \mu \pi + \delta \gamma_u \mu \pi + \gamma_r \gamma_u \mu \omega + \gamma_r \gamma_u \mu \pi + \gamma_r \gamma_u \omega \pi \\
 & + \gamma_r \mu \omega \pi + \gamma_u \mu \omega \pi + \alpha \gamma_a \gamma_r \mu^2 + \alpha \gamma_a \gamma_u \mu^2 + \alpha \gamma_a \mu^2 \omega + \alpha \gamma_a \mu^2 \pi - \alpha \gamma_u \mu^2 \pi - \delta \gamma_r \mu^2 \rho - \delta \gamma_u \mu^2 \rho + \delta \gamma_a \mu^2 \tau \\
 & - \delta \mu^2 \pi \rho + \alpha \gamma_a \gamma_r \gamma_u \mu + \alpha \gamma_a \gamma_r \gamma_u \omega + \alpha \gamma_a \gamma_r \mu \omega + \alpha \gamma_a \gamma_u \mu \omega + \alpha \gamma_a \gamma_r \mu \pi - \alpha \gamma_r \gamma_u \mu \pi + \delta \gamma_a \gamma_r \gamma_u \tau + \alpha \gamma_a \gamma_r \omega \pi \\
 & - \alpha \gamma_r \gamma_u \omega \pi - \delta \gamma_r \gamma_u \mu \rho + \delta \gamma_a \gamma_r \mu \tau + \delta \gamma_a \gamma_u \mu \tau - \delta \gamma_r \gamma_u \pi \rho + \alpha \gamma_a \mu \omega \pi - \alpha \gamma_u \mu \omega \pi + \delta \gamma_a \gamma_r \pi \tau - \delta \gamma_r \gamma_u \pi \tau \\
 & - \delta \gamma_r \mu \pi \rho - \delta \gamma_u \mu \pi \rho + \delta \gamma_a \mu \pi \tau - \delta \gamma_u \mu \pi \tau - \alpha \eta \gamma_r \mu^2 \pi + \alpha \eta \gamma_u \mu^2 \pi - \delta \gamma_a \mu^2 \rho \tau - \alpha \eta \gamma_a \gamma_r \mu \pi + \alpha \eta \gamma_a \gamma_u \mu \pi \\
 & - \alpha \eta \gamma_a \gamma_r \omega \pi + \alpha \eta \gamma_a \gamma_u \omega \pi - \alpha \eta \gamma_r \mu \omega \pi + \alpha \eta \gamma_u \mu \omega \pi - \delta \eta \gamma_a \gamma_r \pi \tau + \delta \eta \gamma_a \gamma_u \pi \tau - \delta \gamma_a \gamma_r \gamma_u \rho \tau - \delta \eta \gamma_r \mu \pi \tau \\
 & + \delta \eta \gamma_u \mu \pi \tau - \delta \gamma_a \gamma_r \mu \rho \tau - \delta \gamma_a \gamma_u \mu \rho \tau - \delta \gamma_a \gamma_r \pi \rho \tau + \delta \gamma_r \gamma_u \pi \rho \tau - \delta \gamma_a \mu \pi \rho \tau + \delta \gamma_u \mu \pi \rho \tau \\
 & + \delta \eta \gamma_a \gamma_r \pi \rho \tau - \delta \eta \gamma_a \gamma_u \pi \rho \tau + \delta \eta \gamma_r \mu \pi \rho \tau - \delta \eta \gamma_u \mu \pi \rho \tau
 \end{aligned} \tag{3.4.3}$$

3.4.2 Robustness to the Choice of Reporting Proportion

Figure 3.7 shows that this hierarchy is **invariant** to the choice of η : across all four panels, the top four parameters remain identical (κ , K , ε , β_e), with elasticity magnitudes essentially unchanged to the third significant digit (for instance, κ moves from -1.0401 at $\eta = 0.05$ to -1.0402 at $\eta = 0.50$). Only lower-ranked parameters – γ_u , γ_r , and η itself – shift slightly in relative position from one panel to the next, an expected mechanical effect since η directly arbitrates the split of cases between the reported (I_r) and unreported (I_u) compartments. The baseline R_c decreases slightly with η (from 5.217 at $\eta = 0.05$ to 5.138 at $\eta = 0.50$), reflecting the fact that a reported case transmits less than an unreported one ($\beta_r < \beta_u$ by construction), but this variation remains minor relative to the magnitude of the dominant elasticities. The quality of epidemiological surveillance, while it influences the magnitude of R_c , therefore does not call into question the identification of the model's structural drivers.

3.4.3 Structural Relevance of the Exposed Compartment E

The second analysis compares the full model (with the exposed compartment E) to the reduced model without E , by quantifying $\Delta = R_c - R_c^e$ and $\Psi = R_c/R_c^e$ through LHS sampling ($n = 3000$, $\pm 20\%$, Figure 3.6).

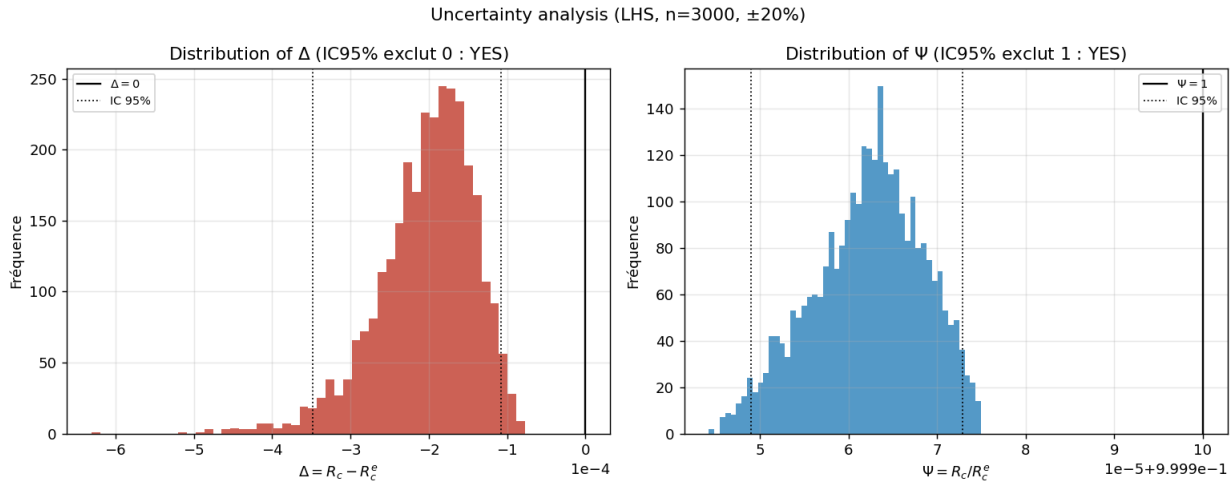


Figure 3.6: Uncertainty distributions of $\Delta = R_c - R_c^e$ (left) and $\Psi = R_c/R_c^e$ (right), LHS $n = 3000$, $\pm 20\%$ around the calibrated values.

Both 95% confidence intervals exclude their reference value ($\Delta = 0$ and $\Psi = 1$, respectively), which, under the strict statistical decision rule adopted, would indicate a structural relevance of E . However, the magnitude of this deviation is negligible: Δ remains bounded within $[-3.5 \times 10^{-4}, -1.1 \times 10^{-4}]$, and Ψ within $[0.99995, 0.99997]$ – an effect on the order of 0.004% of the value of R_c . This dissociation between statistical significance and practical relevance stems from the near-deterministic nature of the underlying relationship: it can be shown analytically that $\Psi = \alpha/(\alpha + \mu)$, a ratio of rates that depends only on the probability of surviving natural mortality during the latent period. With $\alpha \approx 0.71 \text{ day}^{-1}$ (mean incubation ≈ 1.4 days) and $\mu \approx 2.66 \times 10^{-5} \text{ day}^{-1}$ (corresponding to a lifespan of several decades), this ratio is mechanically very close to 1 across the entire perturbation range tested, which explains both the extreme tightness of the two distributions around their asymptotic value and the exclusion of the reference points by the confidence intervals. The compartment E is therefore not structurally determinant for the value of R_c in this model – it nonetheless retains its full relevance for the transient dynamics (realistic incubation delay before infectiousness), which is not the subject of this comparison.

3.5 Sensitivity Analysis and Structural Robustness Results

3.5.1 Agreement Between Local and Global Methods

Two complementary approaches were used to assess the sensitivity of R_c to the model parameters: a local analysis (tornado elasticities, one parameter varied $\pm 20\%$ at a time around the calibrated point) and a global analysis (Latin Hypercube Sampling, $n = 3000$, with Partial Rank Correlation Coefficients, PRCC). The local analysis was additionally repeated for four values of the reporting proportion $\eta \in \{0.05, 0.20, 0.35, 0.50\}$, to verify that the ranking of influential parameters does not depend on an arbitrary assumption about the quality of epidemiological surveillance (Figure 3.7).

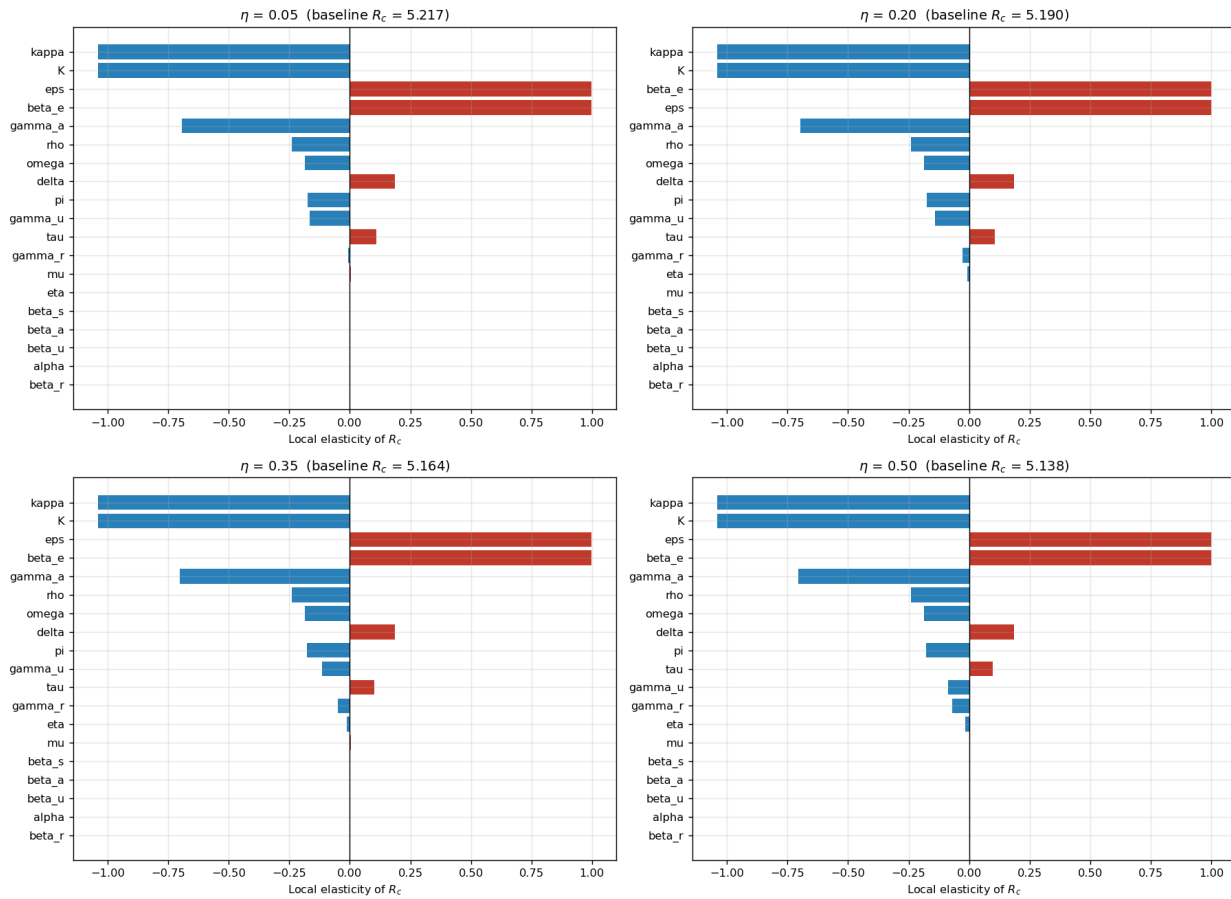


Figure 3.7: Local elasticities of R_c (tornado analysis, $\pm 20\%$) for four values of the reporting proportion η .

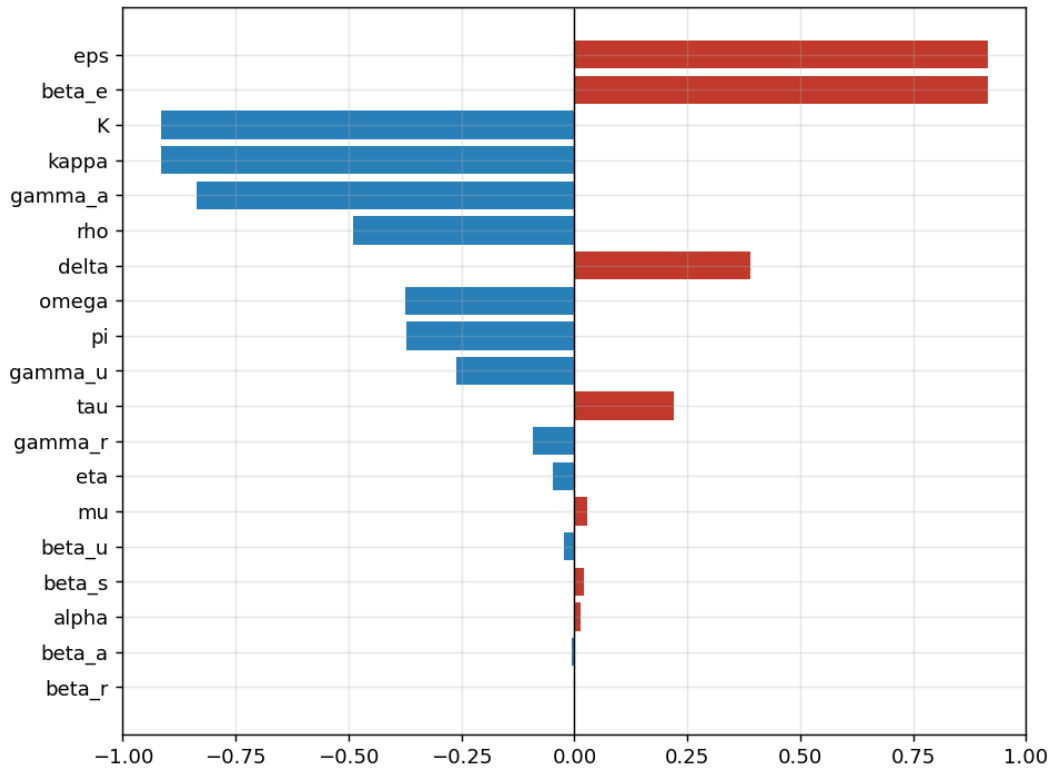


Figure 3.8: Partial Rank Correlation Coefficients (PRCC) of R_c with respect to the model parameters (LHS, $n = 3000$, $\pm 20\%$).

Both methods converge on the same hierarchy, which is itself a methodological robustness result: the environmental parameters ε (bacterial shedding rate) and β_e (environmental transmission coefficient) dominate, with elasticity and PRCC values close to $+1$ and $+0.9$ respectively, while K (half-saturation constant) and κ (bacterial decay rate) are symmetrically opposed (elasticity ≈ -1.04 , PRCC ≈ -0.9). This quadruplet $(\varepsilon, \beta_e, K, \kappa)$ alone accounts for the vast majority of the uncertainty in R_c , consistent with the route decomposition of R_c obtained at the previous step, where the environmental pathway R_e carried more than 99% of the total value.

The recovery rate of asymptomatic individuals γ_a occupies an intermediate but notable position (elasticity ≈ -0.70 , PRCC ≈ -0.85), followed by a group of moderately influential parameters ($\rho, \delta, \omega, \pi, \gamma_u, \tau$, all between 0.1 and 0.5 in absolute value) governing the distribution of individuals across the infectious and vaccinated classes. Finally, the human transmission parameters calibrated on data ($\beta_s, \beta_a, \beta_r, \beta_u$), together with α, μ, η , and γ_r , show near-zero elasticity and PRCC in both analyses – further confirmation that the calibration procedure does not introduce artificial instability into the model, and that the sensitivity of R_c is carried almost entirely by the environmental subsystem rather than by human-to-human transmission.

3.5.2 Summary

Taken together, these three analyses paint a coherent picture of the calibrated model: (i) the sensitivity of R_c is concentrated on the environmental subsystem $(\varepsilon, \beta_e, K, \kappa)$ and on the recovery of asymptomatic individuals (γ_a) , a result consistent across local and global methods and robust to the choice of η ; (ii) the data-calibrated human transmission parameters introduce no excessive sensitivity, indirectly validating the calibration procedure; and (iii) the inclusion of the exposed compartment E , although statistically detectable, has no practical bearing on the epidemic threshold and can be retained purely for its dynamical realism without risk of biasing the estimation of R_c .

4. Discussion

Taken together, the results of Chapter 3 paint a picture of the SEIR-B model that is more structurally settled than the operational picture one might have expected *a priori*: the bifurcation at $R_c = 1$ is robustly forward, the exposed compartment E turns out to be statistically detectable but practically inconsequential for the value of R_c , and the sensitivity of R_c is concentrated almost entirely on a small block of environmental parameters rather than on the human-side, data-calibrated transmission coefficients. We discuss each of these findings in turn, then draw out their operational implications.

The center-manifold analysis (Section 3.2), confirmed numerically by branch continuation, Jacobian spectral analysis, and phase-portrait inspection (Section 3.3), shows a classical forward transcritical bifurcation at $R_c = 1$: the coefficient $a(\varphi)$ remains strictly negative over the entire physically admissible range of the natural immunity waning rate φ , from a 30-day to a 20-year mean duration of protection (Table 3.1), and the critical threshold φ_c at which a backward regime could in principle appear is found to be negative ($\varphi_c \approx -3.79 \times 10^{-3} \text{ day}^{-1}$), i.e. outside the biologically meaningful domain $\varphi \geq 0$. No bistability window was found in which a stable endemic equilibrium would coexist with a stable disease-free equilibrium for $R_c < 1$. This is a reassuring and decision-relevant result: within this model, driving R_c below unity, through vaccination, sanitation, or surveillance, alone or in combination is both necessary *and sufficient* for elimination. Unlike models exhibiting backward bifurcations, there is no additional “overshoot” effort required, nor any risk that the disease persists despite $R_c < 1$ because of a hidden bistable regime. The damped, oscillatory approach to the endemic equilibrium observed for $R_c > 1$ (Figures 3.4–3.5) is consistent with a pair of complex eigenvalues with negative real part near that equilibrium, and is not indicative of a Hopf bifurcation, since the real part of that pair remains strictly negative throughout the explored range.

The comparison strategy of Section 2.3, applied to the calibrated parameter set, shows that including E shifts R_c by $\Delta = R_c - R_c^e \in [-3.5 \times 10^{-4}, -1.1 \times 10^{-4}]$ and $\Psi = R_c/R_c^e \in [0.99995, 0.99997]$ (95% CI, LHS, $\pm 20\%$), both intervals exclude their respective null values (0 and 1), which, under the decision rule stated in Section 2.3, formally indicates a structurally detectable effect of E . However, the *magnitude* of this effect is on the order of 0.004% of R_c , several orders of magnitude below any threshold of practical relevance. This apparent tension between statistical significance and practical irrelevance is fully explained by the closed-form identity $\Psi = \alpha/(\alpha + \mu)$ derived in Section 3.4.3: the entire effect of E on R_c operates through a single mechanism, namely the probability of dying of natural causes *during* the latent period, before becoming infectious. Since the mean incubation period ($1/\alpha \approx 1.4$ days) is more than four orders of magnitude shorter than the

mean human lifespan implied by μ , this probability is mechanically close to zero regardless of the parameter perturbations explored, so that Ψ is squeezed into an extremely narrow band close to 1, narrow enough that even a negligible, near-deterministic shift is enough to exclude the null from the confidence interval. This is consistent with, and refines, earlier findings in the literature that latency reduces net transmissibility through pre-infectious losses [23, 24, 25]: here we show *quantitatively* that, for a disease with an incubation period as short as cholera's relative to the host lifespan, this reduction is real but negligible. The practical conclusion is not that E is dispensable, it remains essential for realistically capturing the delay between exposure and infectiousness, and therefore the timing and shape of simulated outbreaks, but that its *omission would not materially bias the estimated epidemic threshold* in this setting, contrary to what a purely local, single-point sensitivity check might have suggested.

The five-route decomposition of proposition 3.1.10 and its numerical evaluation (Table 3.2) show that, at the calibrated parameter point, $R_c \approx 5.17$ is carried at 99.9% by the environmental route R_e , with the four human-to-human routes (R_s, R_a, R_r, R_u) summing to approximately 0.008. Both the local (tornado) and global (PRCC) sensitivity analyses corroborate this decomposition and agree closely with one another: the environmental block ($\varepsilon, \beta_e, K, \kappa$) dominates, with elasticities and PRCC values near ± 1 and ± 0.9 respectively, followed at some distance by the asymptomatic recovery rate γ_a (elasticity ≈ -0.70 , PRCC ≈ -0.85), a group of moderately influential demographic and progression parameters ($\rho, \delta, \omega, \pi, \gamma_u, \tau$), and finally the calibrated human transmission coefficients ($\beta_s, \beta_a, \beta_r, \beta_u$) together with $\alpha, \mu, \eta, \gamma_r$, all of which show near-zero influence on R_c in both analyses. This hierarchy is essentially unchanged across the four reporting-proportion scenarios $\eta \in \{0.05, 0.20, 0.35, 0.50\}$ (Figure 3.7): only lower-ranked parameters directly tied to the reporting split (γ_u, γ_r, η itself) move slightly between panels, a mechanical consequence of how η redistributes cases between I_r and I_u , and the baseline R_c itself only drifts modestly (from 5.217 to 5.138) over this range. Structurally, this is consistent with the foundational finding of Codeço [2] that endemic cholera cannot be explained without an aquatic reservoir, and with subsequent work showing that hydrologic and seasonal conditions can substantially amplify the environmental contribution to transmission [28, 29, 30]. It should be stressed, however, that ε, κ , and K were not estimated from the calibration data, they are fixed at literature-anchored values, for reasons of structural identifiability discussed in Section 2.1.4 so that their dominance in the sensitivity analysis reflects the width of their assumed prior ranges as much as an intrinsic property of the fitted model. The message is nonetheless useful: *given the uncertainty we currently have* about the environmental parameters, this is where that uncertainty most strongly propagates into R_c , and where additional field data (environmental *V. cholerae* monitoring) would be most valuable.

These three results together reframe the operational reading of the model relative to a purely local, single-scenario analysis. First, because the human transmission parameters, including those governing detection and isolation ($\eta, \pi, \gamma_u, \beta_u$), carry near-zero elasticity and PRCC on R_c in the calibrated regime, our results do *not* support the expectation that strengthening case detection and notification alone would, by itself, produce a large reduction in R_c : its main epidemiological value in this model lies in improving the accuracy of burden estimation and enabling faster, better-targeted outbreak response, rather than in shifting the reproduction number materially. Second, because R_c is overwhelmingly carried by the environmental route, interventions that act on that route; Water, Sanitation and Hygiene (WASH) infrastructure, safe water access, and household or community water treatment, which respectively influence β_e, K , and effectively ε and κ are, on the basis of this sensitivity ranking, the highest-yield structural levers for reducing R_c , corroborating long-standing arguments for the centrality of WASH in cholera control [2, 28, 30], particularly ahead of and during flood-prone seasons. Third, maintaining population immunity through oral cholera vaccination (coverage ω , efficacy ϱ , slow waning δ) remains a relevant complementary lever, consistent with its moderate-but-nonzero ranking in both sensitivity analyses. Finally, because no backward-bifurcation regime was found, none of these interventions need to *overshoot* the $R_c = 1$ threshold to guarantee elimination: bringing R_c below 1 is, within this model, sufficient on its own.

Several limitations qualify these conclusions. The calibration target is a constructed weekly proxy series consistent with real WHO aggregate totals for West Africa in 2024, not an observed spatiotemporal surveillance series (Section 2.1.2); direct calibration to a real linelist or weekly reporting series, when available, would sharpen the point estimates of the human transmission parameters, though it is unlikely to alter the qualitative conclusion that the environmental route dominates, since that conclusion is driven by the fixed environmental parameters rather than by the calibrated ones. The global sensitivity analysis assumes independent marginal distributions for all parameters under Latin Hypercube Sampling; plausible correlations, for instance between η, π , and γ_u (all linked to the same detection and case-management pathway), were not modelled and could reshape the PRCC ranking among that specific cluster of parameters, though they are unlikely to displace the environmental block from the top of the hierarchy given the size of the gap observed. Finally, the model does not include explicit seasonal forcing or spatial structure, both of which are known to modulate the relative weight of the environmental pathway [28, 29]. These limitations point directly to natural extensions. In particular, the three main findings of this thesis, the forward bifurcation, the negligible structural role of E , and the environmental dominance of R_c currently follow from the model's structure combined with literature-anchored parameter values and a synthetic calibration target, rather than from direct empirical validation. Hierarchical or Bayesian calibration against

real spatiotemporal surveillance data, together with seasonal or spatially explicit scenarios, would make it possible to test whether these three conclusions still hold once real data are used, instead of relying on the model's current assumptions to infer that they do.

In summary, explicitly separating reported and unreported cases, and retaining the exposed compartment for dynamical realism even though it is not decisive for the epidemic threshold, together with a rigorous, literature-anchored treatment of the environmental parameters, yields a model whose qualitative conclusions are robust across detection scenarios and across the explored parameter space. The clearest actionable insight is that, under the uncertainty currently quantifiable from available data, environmental and WASH-oriented interventions constitute the primary lever on transmissibility in this West African cholera setting, with vaccination and case management as essential complements rather than the principal drivers of R_c .

5. Conclusion

This thesis developed and analyzed a structurally disaggregated SEIR-B model for cholera transmission in West Africa, combining an explicit latent (exposed) stage, a bacterial environmental reservoir, and a four-way split of infectious individuals into asymptomatic, symptomatic, reported, and unreported classes. Three complementary strands of analysis, analytical (existence, positivity, boundedness, equilibrium and stability theory, and a closed-form next-generation-matrix decomposition of R_c), numerical (bifurcation continuation, calibration on real WHO and UN data), and statistical (local and global sensitivity analysis) were brought together to assess not only the value of the control reproduction number R_c , but the *robustness* of that value and of the policy conclusions drawn from it.

Four main results emerge from this work.

First, the disease-free equilibrium was shown to be locally and, under an explicit and interpretable invariance condition, globally asymptotically stable whenever $R_c < 1$, and the bifurcation occurring at $R_c = 1$ was established analytically, via the center manifold theorem, and confirmed independently by three numerical methods (branch continuation, Jacobian spectral analysis, and phase-portrait inspection) to be **forward** across the entire physically plausible range of the natural immunity waning rate. No backward-bifurcation regime, and therefore no bistability or elimination “overshoot” requirement, was found: reducing R_c below unity is both necessary and sufficient for disease elimination in this model.

Second, the structural role of the exposed compartment E was quantified analytically for the first time in this framework through the closed-form identity $\Psi = R_c/R_c^e = \alpha/(\alpha + \mu)$, and confirmed numerically by Latin Hypercube Sampling. Although the 95% confidence intervals of $\Delta = R_c - R_c^e$ and Ψ formally exclude their null values, the magnitude of the effect is negligible ($\approx 0.004\%$ of R_c), because the incubation period is many orders of magnitude shorter than the human lifespan. Explicitly modelling E is therefore not required to obtain an accurate estimate of the epidemic threshold in this setting, although it remains valuable for capturing realistic transient, time-to-infectiousness dynamics.

Third, the five-route decomposition of R_c , combined with local (tornado) and global (PRCC) sensitivity analyses conducted across four detection scenarios ($\eta \in \{0.05, 0.20, 0.35, 0.50\}$), showed that $R_c \approx 5.17$ is carried almost entirely (99.9%) by the environmental transmission route, and that this dominance, along with the underlying parameter ranking, is robust to the assumed level of case detection. The data-calibrated human transmission coefficients consistently ranked among the least influential parameters in both analyses, indicating that the calibration procedure introduces no artificial instability and that, under current data

and parameter uncertainty, the environmental pathway, not human-to-human contact, is the primary determinant of transmissibility in this setting.

Fourth, and consequently, the operational reading of the model departs from a case-detection-centred strategy: because η , π , γ_u , and β_u carry negligible influence on R_c in the calibrated regime, WASH-oriented interventions acting on the environmental parameters (β_e , K , and effectively ε , κ) emerge as the highest-yield structural levers, with sustained vaccination coverage as an important complement and improved detection remaining valuable chiefly for surveillance accuracy and response speed rather than for its direct effect on R_c .

These findings are naturally bounded by the scope of the data and assumptions used. The calibration relies on a constructed weekly proxy series anchored to real annual WHO totals rather than an observed spatiotemporal surveillance series; the environmental parameters ε , κ , and K are fixed at literature-anchored values rather than estimated, for reasons of structural identifiability; the global sensitivity analysis assumes independent parameter marginals; and the model does not yet incorporate explicit seasonal forcing or spatial structure, both known to modulate the environmental contribution to cholera transmission. Addressing these limitations defines a clear path for future work: (i) hierarchical or Bayesian calibration against real spatiotemporal surveillance data, ideally including direct environmental *V. cholerae* measurements to relax the fixed-parameter assumption on ε , κ , K ; (ii) seasonal and hydrologically forced extensions with spatial or mobility-informed coupling; (iii) stochastic formulations better suited to capturing early-outbreak variability and extinction risk; and (iv) a formal portfolio-optimization layer combining the epidemiological thresholds derived here with intervention costs, to translate the sensitivity ranking into an explicit resource-allocation recommendation.

In summary, this work provides a decision-relevant, structurally robust assessment of cholera transmission risk in West Africa: the epidemic threshold behaves predictably (forward bifurcation, no hidden bistability), the exposed stage can be retained for realism without distorting that threshold, and under present data and parameter uncertainty environmental exposure, not case detection, is the dominant lever on transmissibility. Control portfolios that prioritize WASH investment, sustain vaccination coverage, and use case detection primarily to guide and monitor these efforts are best supported by the present analysis, and should be validated and refined as local, seasonally resolved surveillance and environmental data become available.

Acknowledgements

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